

# Bayesian optimisation of compressive properties in particulate composites based on a bio-sourceable unsaturated polyester resin

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Thomas Højlund-Dodd, Mike Bennison, Danny K. H. Ho, James Elliott

Department of Materials Science and Metallurgy, University of Cambridge, Charles Babbage Road, Cambridge, CB3 0FS, UK

## Abstract

The mechanical performance of itaconic-acid-derived unsaturated polyester composites depends on the coupled effects of polyester, reactive diluent, initiator and particulate-filler contents. Here, Bayesian optimisation was used to efficiently improve the compressive mechanical properties of a calcium-sulphate-filled bio-sourceable resin based on poly(propylene itaconate) (PPI) and dimethyl itaconate. Three formulation variables were explored using eight initial Sobol samples followed by 16 sequential experiments. Separate single-objective routines targeted apparent compressive modulus and peak compressive stress and were benchmarked against a sequential space-filling MIPT (mc-intersite-proj-th) sampler. Within the 24-sample budget, Bayesian optimisation identified formulations exhibiting an apparent compressive modulus of 5,012 MPa and a peak compressive stress of 143 MPa. The corresponding MIPT maxima were 4,851 MPa and 140 MPa, respectively. An otherwise identical styrene-diluted analogue reached 7,271 MPa, such that the matched-budget dimethyl-itaconate system retained 69% of its modulus. A subsequent multi-objective refinement, initialised using the combined 56-sample dataset, identified a formulation exhibiting 5,458 MPa and 165 MPa, simultaneously exceeding the initial empirical Pareto set. These results demonstrate the value of sequential optimisation for navigating coupled formulation variables in emerging unsaturated polyester composite systems.

## 1 Introduction

Thermoset composites underpin structural applications ranging from construction panels to transport components due to their high stiffness, dimensional stability and chemical resistance<sup>1</sup>. However, unlike thermoplastics, crosslinked thermosets cannot be readily remelted or reshaped, and end-of-life options remain dominated by incineration and landfill<sup>2</sup>. In 2019 alone, 353 Mt of plastic waste was generated globally, of which only 9 % was recycled, while thermoset-rich composite waste is increasingly directed toward energy recovery through combustion<sup>2</sup>. As most commercial thermosets are derived from fossil feedstocks, their incineration rein-

troduces fossil carbon into the atmosphere, directly coupling materials performance to carbon-cycle perturbation<sup>3</sup>.

Since the vast majority of the virgin plastic manufactured annually is thermoplastic, this group has naturally dominated the polymer sustainability literature, with papers focused upon improvement of recycling methods<sup>4</sup>, mitigation of microplastic waste<sup>5</sup>, and biodegradability<sup>6</sup>. However, in part due to their relatively low industrial volume, thermosets have received far less attention<sup>7</sup>. It is generally impractical to depolymerise crosslinked thermoset materials which has lead to the widespread landfilling or incinerating of these materials at end-of-life<sup>8-10</sup>. Contemporary European legislation has prioritised

incineration ahead of landfill for composite materials with thermosetting matrices, and recent legislation (e.g. Directives ‘99/31/EC’ and ‘2000/53/EC’) is likely to increase the proportion incinerated<sup>11</sup>. As the majority of thermoset plastics are derived from fossil-based sources of carbon, their combustion releases the geologically sequestered carbon into the atmosphere, resulting in a net addition of CO<sub>2</sub> into the short-term carbon cycle<sup>12</sup>. The present work addresses this challenge by initially identifying a suitable bio-based thermoset chemistry, whereafter it seeks to efficiently optimise their formulation for compressive performance. Here, fully bio-sourceable denotes a resin whose organic constituents can all be produced from biogenic feedstocks using established or scalable routes. In this study, the commercial reagent batches were not independently certified as biogenic.

### Bayesian Optimisation of Thermosets

In polymer science, Bayesian optimisation has been deployed in both purely computational domains (e.g. optimisation of thermoplastic repeat units for improved glass transition temperatures in a simulated parameter space<sup>13</sup>) as well as in applied practice (e.g. improving compositional homogeneity during co-polymerisation of styrene and methyl methacrylate<sup>14</sup>). Knox et al. applied autonomous multi-objective Bayesian optimisation to RAFT polymerisation, coupling a flow reactor with online NMR and at-line GPC to map the Pareto trade-off between monomer conversion and molar-mass dispersity<sup>15</sup>. Various thermosets have been practically optimised using Bayesian optimisation: benzoxazine-based thermosets have been optimised with respect to six thermomechanical properties<sup>16</sup>, polycyanurates have had three thermal/mechanical/hygroscopic properties improved upon<sup>17</sup>, and epoxy thermosets based on amine curing agents have been optimised with respect to a trio of mechanical properties using a dual-driver method<sup>18</sup>. In the more specific domain of

practical Bayesian optimisation of bio-based thermosets, literature has been dominated by epoxy-based chemistries; Albuquerque et al. successfully improved glass transition temperatures in bio-based epoxy thermosets derived from amino acid based resin systems<sup>19</sup>, whilst Rothenhäusler et al. explored this class of thermosets further by taking into account both cost and bio-based content<sup>20</sup>.

Bayesian optimisation has seen practical application in the development of thermoset composites with a paper by Schoenholz et al. considering unidirectional carbon fibre prepreg based on epoxy resin. In that work, both the layup and curing cycle variables were optimised to reduce the process-induced deformation in final composite parts<sup>21</sup>. Despite this, relatively little literature exists concerning the practical application of Bayesian optimisation in the development of properties in polyester-based thermosets, their bio-based analogues, or their derived composites. In applying Bayesian optimisation for the development of strength in simple composites based on itaconate-derived thermoset chemistries, this paper addresses this gap in the literature.

### Bio-Based Thermoset Chemistry

Thermosets are formed from a wide range of different chemistries including (but not limited to): phenolic resins, unsaturated polyester resins, epoxy resins, and polyurethanes<sup>22</sup>. One prominent group are the unsaturated polyester resins, which are routinely used in the manufacture of composite components. These resins exhibit desirable properties including ease of handling in liquid form, rapid curing, and excellent dimensional stability<sup>23,24</sup>. An archetypal unsaturated polyester resin is the mixture of poly(propylene fumarate) copolymer (polymerised from propylene glycol and maleic anhydride) with a styrene reactive diluent<sup>25</sup>. This conventional formulation is currently manufactured from fossil feedstocks. For example, propylene glycol is industri-

ally manufactured from propylene oxide, a downstream chemical product of propane extracted from natural gas<sup>12</sup>. Maleic anhydride is synthesised by catalytic oxidation of the chemical butane<sup>26</sup>, this derived from fractional distillation of petroleum crude<sup>27</sup>. Furthermore, styrene is produced from the dehydrogenation of ethylbenzene- this being the product of a reaction between benzene and ethylene, both of which are fossil-fuel derived feedstocks<sup>12,28</sup>.

Although the literature describes numerous attempts at synthesising the aforementioned conventional monomers from bio-based feedstocks (e.g. maleic anhydride<sup>29–31</sup> and styrene<sup>32</sup>) only research into propylene glycol<sup>33–36</sup> has achieved apparent success in scale-up and commercialisation, so we retained this monomer in the present work. Also incorporated into this work polyester synthesis was itaconic acid, a simple bio-based difunctional carboxylic acid manufactured by fungal fermentation of sugars by fungi such as *Aspergillus terreus*<sup>37</sup>. Itaconic acid’s dicarboxyl and monoalkene functionality allows it to directly substitute maleic acid during polycondensation to form an unsaturated polyester<sup>38</sup>. Furthermore, itaconic acid can be synthesised into a variety of diesters (the family of diitaconates) which exhibit properties appropriate of a reactive diluent (i.e. radical initiated homopolymerisation and a low viscosity), allowing them to replace styrene in an eventual unsaturated polyester resin<sup>39</sup>. As an aside, the use of methanol in this work to synthesise dimethyl itaconate was justified by the fact that a known route to bio-derived methanol exists<sup>40</sup> and that commercial scale-up has been publicly acknowledged.

Itaconic acid’s potential is clear in the literature, where subjects including the catalytic influence on polycondensation of itaconic acid within eventual unsaturated polyesters<sup>41</sup>, synthesis of partially bio-based resins incorporating itaconic-acid-based unsaturated polyester but retaining styrene as the reactive diluent<sup>42</sup>, as well as fully

bio-based resins using itaconic-acid-based unsaturated polyester in tandem with diitaconates as reactive diluents<sup>43,44</sup> have been disseminated. Furthermore, previous studies have considered partially bio-based resins incorporating itaconic-acid-based unsaturated polyesters together with the cross-linking compatible methacrylate class of reactive diluents<sup>45,46</sup>. Although conventionally fossil-fuel derived, recent work has seen bio-based methyl methacrylate synthesised from itaconic acid, which would unlock fully bio-based resins based on methacrylate reactive diluents in future<sup>47,48</sup>. Despite inferior homopolymerisation rates<sup>39</sup> and viscosity characteristics<sup>45</sup> relative to methacrylates, diitaconates were nonetheless focused upon in this research ahead of methacrylates on account of their relatively higher strengths as cured thermosets<sup>45</sup>, safety as reactive diluents (lower toxicity and flammability)<sup>49</sup>, and ease of handling on account of their lower volatility<sup>45,50</sup>.

## A Thermoset Particulate Composite

In 1997, Kharas et al. prepared several samples of bioresorbable bone cement composites using both polyester-based thermosets in combination with particulate calcium sulphate, where the latter’s content was found to affect the final thermosets’ strength and modulus<sup>51</sup>. The sensitivity of mechanical properties toward this variable inspired the present work’s adoption of anhydrous calcium sulphate as a particulate reinforcing element during the preparation of simple particulate composites based on a fully bio-sourceable unsaturated polyester resin.

As a counterpart to this particulate reinforcing, the thermosetting matrix was based on a cross-linked unsaturated polyester system; these being reliant upon a combination of both condensation and addition polymer. The condensation polymer is typically an unsaturated linear polyester formed from the copolymerisation of diols and diacids, with at least one monomer having an unsaturated double bond in its structure

which is retained after copolymerisation- this copolymer is referred to as an unsaturated polyester<sup>52</sup>. Whilst the addition polymer is typically generated through a radical-addition homopolymerisation reaction, where a monomer which had been acting as an unsaturated polyester solvent reacts and crosslinks individual unsaturated polyester chains- these monomers are therefore referred to as reactive diluents<sup>53</sup>. The combination of unsaturated polyester and reactive diluent is known as an unsaturated polyester resin; a viscous liquid, which, on addition of radical-initiating chemicals, can rapidly form a rigid thermoset material (i.e. the composite’s thermoset matrix). Previous work by Panic et al. found dimethyl itaconate to be the strongest contender of the diitaconates as a bio-based replacement for styrene in unsaturated polyester resins, given that resultant cured thermosets exhibited relatively higher crosslink densities (73% the density of a styrene-based thermoset)<sup>44</sup>. This work therefore sought to replicate these unsaturated polyester resins, where the reactive diluent was pure dimethyl itaconate and the unsaturated polyester was poly(propylene itaconate).

Having both described the particular material considered in the present work and provided a thorough review of Bayesian optimisation’s utility in the sample-efficient formulation of thermoset materials, the paper will take the following form: Initially, a detailed description of experimental methods is provided, spanning feedstock synthesis, bio-sourceable unsaturated polyester resin synthesis, and particulate composite preparation, followed by a discussion of the optimisation procedures implemented. Subsequently, a results and discussion section summarises the findings arising from both the single-objective and dual-objective routines undertaken, as well as a comparison with the optimisation of a model conventional resin based on styrene. Finally, we conclude by noting the most salient findings, including the successful single-objective Bayesian

optimisation of these materials to strengths approaching those of styrene analogues.

## 2 Materials & Methods

Optimisation experiments carried out here involved both the practical methods associated with preparation and characterisation of thermoset composite samples as well as computational techniques developed to carry out Bayesian optimisation based on data obtained. Furthermore, synthetic routes are presented for producing any materials not readily available commercially. This section will first present practical methods including synthesis of itaconic acid polyester and diesters, thermoset composite preparation, and characterisation techniques employed. Subsequently, the computational method by which Bayesian optimisation was implemented will be elucidated.

### 2.1 Materials

Propylene glycol (99%), itaconic acid (99%), anhydrous calcium sulphate (99%), and hydroquinone (99%) were obtained from “Thermo Scientific Chemicals”, a division of *Thermo Fisher Scientific Inc.* based in Waltham, Massachusetts, U.S. Toluene (99.8%), dried magnesium sulphate (99%), sodium chloride (99.5%), and dichloromethane (95%) were obtained from “Fisher Chemicals”, a division of *Thermo Fisher Scientific Inc.* Mixed xylenes (97%) were obtained from “Alfa Aesar” a division of *Thermo Fisher Scientific Inc.* Methanol (99.9%) was obtained from “Honeywell Riedel-de Haën”, a brand under *Honeywell International Inc.* based in Charlotte, North Carolina, U.S. P-toluenesulphonic acid monohydrate (98%) and chloroform-d (99.8%) were obtained from “Sigma-Aldrich”, a subsidiary of *Merck KGaA* based in Darmstadt, Germany. Dibenzoyl peroxide (20%) filled with calcium sulphate was obtained from *Nouryon Specialty Chemicals B.V.* based in Amsterdam, The Netherlands.



Sodium bicarbonate (99%) was obtained from *Fluorochem Ltd* based in Glossop, Derbyshire, U.K. Phenolphthalein solution (1% w/w in ethanol) and potassium hydroxide (0.56% w/w in methanol) were supplied by *Scientific Laboratory Supplies Ltd* based in the East Riding of Yorkshire, U.K.

## 2.2 Diitaconate Synthesis & Characterisation

Dimethyl itaconate was synthesised by esterification of itaconic acid with methanol by the Fischer method which has previously been used successfully in the literature<sup>50,54,55</sup>; this is presented in Figure 1 as a simplified reaction scheme. The method was adapted from the work of Trejbal et al. and involved the mixing of itaconic acid in an excess of methanol in a molar ratio of 1:3<sup>50</sup>, whilst p-toluenesulphonic acid was used as a strong acid catalyst and made up 4% w/w of the final reaction mixture. Detailed practical method by which this was achieved has been placed in the supporting information (Sec. 9.1), but may be shortly summarised: initially the reaction mixture is refluxed to produce the impure product, liquid-liquid extraction is then employed to draw out the pure product into a suitable organic solvent, and finally this solvent is removed under reduced pressure to yield the final product.

Several different characterisation techniques were used to monitor the reaction progress, identify chemicals, and ascertain the product purity, including thin layer chromatography (TLC), Fourier-transform infrared spectroscopy using attenuated total reflectance (ATR-FTIR), and <sup>1</sup>H and <sup>13</sup>C nuclear magnetic resonance spectroscopy (NMR). Preliminary investigations undertaken using TLC have been detailed in the supporting information (see Sec. 9.2) for the sake of brevity.

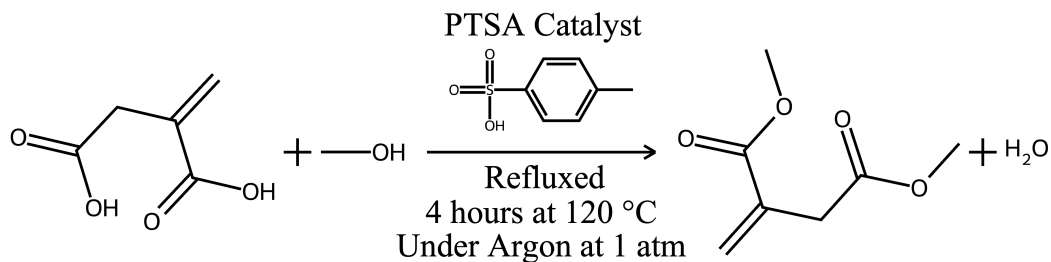
ATR-FTIR spectra were obtained using a Nicolet iS50 machine with the spectral range between 400-4000 cm<sup>-1</sup> being investigated at a resolution of 0.09 cm<sup>-1</sup>. Figure 2

presents the spectra obtained for the pure reactant (itaconic acid) and compares it with that of the product isolated after rotary evaporation. These data support the hypothesis that the reactant was successfully esterified to form the product dimethyl itaconate in a number of ways.

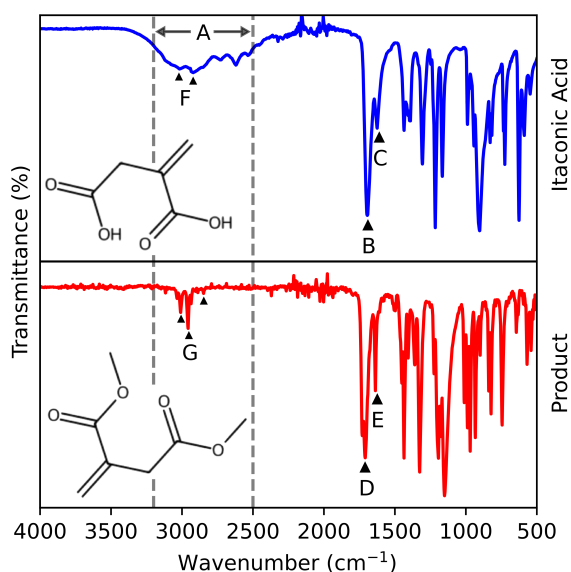
Firstly, the broad peak characteristic of O—H bond stretching commonly located between 2500-3200 cm<sup>-1</sup> is clearly present for the itaconic acid but not for the product (Fig. 2A), suggesting the loss of O—H bonds and therewith the carboxyl groups associated with either the initial reactant (itaconic acid) or the intermediate product (monomethyl itaconate).

Secondly, strong peaks are observed around 1710 cm<sup>-1</sup> (Fig. 2B & 2D) which is commonly indicative of C=O bond stretching (which occurs between 1640-1760 cm<sup>-1</sup>); this would be consistent with C=O bond retention which would occur during successful esterification. Furthermore, it may be noted that peaks associated with carboxylic C=O bonds occur between 1640-1730 cm<sup>-1</sup> whilst ester C=O peaks may occur between 1690-1750 cm<sup>-1</sup>; the positive difference between the reactant C=O peak located at 1700 cm<sup>-1</sup> (Fig. 2B) and the product's peak at 1720 cm<sup>-1</sup> (Fig. 2D) would support successful esterification.

Thirdly, an intermediate-sized peak is observed both for the reactant (Fig. 2C) and the product (Fig. 2E) at 1630 and 1640 cm<sup>-1</sup> respectively, which is suggestive of nonconjugated C=C bond stretching (occurring between 1620—1680 cm<sup>-1</sup>), which would be consistent with retention of the double bond during the proposed esterification. As an aside, clusters of peaks likely associated with C-H bond stretching were observed on both spectra (Fig. 2F & 2G) but inter-comparison was obfuscated by the strong and broad O—H stretching signal apparent in the reactant's spectrum.



**Figure 1:** Simple reaction scheme for synthesis of dimethyl itaconate.



**Figure 2:** ATR-FTIR spectra of reactant and product. (A) O—H stretch at 2500—3200  $\text{cm}^{-1}$ . (B) C=O stretch at around 1700  $\text{cm}^{-1}$ . (C) 1630  $\text{cm}^{-1}$ . (D) C=O stretch at around 1720  $\text{cm}^{-1}$ . (E) 1640  $\text{cm}^{-1}$ . (F) 2950 & 3050  $\text{cm}^{-1}$ . (G) 2875, 2980 & 3035  $\text{cm}^{-1}$ .

Both  $^1\text{H}$  and  $^{13}\text{C}$  NMR spectroscopy was carried out on a sample of product using a Bruker Avance III HD 500 MHz Smart Probe spectrometer, with spectra produced at 500 MHz and 125 MHz respectively. Chemical shifts are considered as downfield shifts ( $\delta$  in ppm) from the tetramethyl silane standard ( $\delta = 0$  ppm) and calibrated using the shift exhibited by trace amounts of non-deuterated chloroform ( $\delta = 7.26$  ppm singlet for  $^1\text{H}$  and  $\delta = 77$  ppm triplet for  $^{13}\text{C}$ ) within the sample solvent. The chloroform-d solvent also introduced a trace amount of water which exhibited a singlet peak ( $\delta = 1.63$  ppm) in the  $^1\text{H}$  spectrum; these peaks were ignored when examining the spectra for dimethyl itaconate, as presented here:

Dimethyl itaconate:  $^1\text{H}$  NMR ( $\text{CDCl}_3$ , 500 MHz):  $\delta$  3.34 (dd, 2H,  $J = 0.71, 0.15$  Hz), 3.69 (s, 3H), 3.76 (s, 3H), 5.71 (q, 1H,  $J = 0.97$  Hz), 6.32 (dt, 1H,  $J = 0.84, \text{n/a}$  Hz).  $^{13}\text{C}\{^1\text{H}\}$  NMR ( $\text{CDCl}_3$ , 125 MHz):  $\delta$  37.5, 52.1, 52.2, 128.6, 133.7, 166.6, 171.1.

Note, n/a was used as a placeholder for the smaller of the two coupling constants exhibited by a suspected doublet of triplets at 6.32 ppm, this was on account of long range allylic couplings not being well enough resolved to ascertain this value. A quantitative  $^1\text{H}$  spectrum showed water constituting a 5% w/w impurity, although the dimethyl itaconate sample purity is expected to exceed 95% given that the deuterated solvent will have introduced an unknown amount of water. This proved that the synthetic method devised was capable of producing dimethyl itaconate at a technical grade or above in terms of purity, allowing the subsequent experimental work to proceed.

### 2.3 Unsaturated Polyester Resin Synthesis & Characterisation

Unsaturated polyester was synthesised via the polyesterification of itaconic acid with propylene glycol to produce poly(propylene itaconate), as described by the simplified reaction scheme in Figure 3. This was achieved using the melt polycondensation method, since a precedent exists in the literature for its successful use in the synthesis of linear unsaturated polyesters derived from itaconic

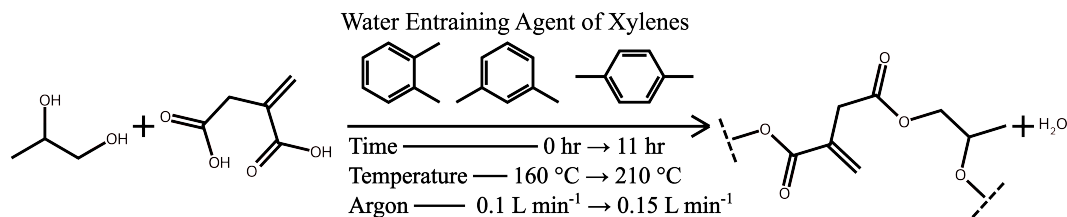
acid<sup>43–46</sup>. The stoichiometric ratio between itaconic acid and propylene glycol was set at 1:1.1, on the basis that propylene glycol is lost during the heating process which reaches temperatures exceeding its boiling point<sup>56</sup>.

Initially, the ‘azeotropic method’ of melt polycondensation was undertaken whereby a reaction vessel was charged with diols, diacids, and a water-entraining agent (5% w/w mixed xylenes) before being heated for an extended period (approximately 11 h). Subsequently, an inhibiting agent (0.06% w/w hydroquinone) was added to the molten polyester before combination with reactive diluent (30% w/w dimethyl itaconate) to produce the final unsaturated polyester resin. A more detailed description of this method may be found in the supporting information under Section 9.3. As a predictor variable which was to be used during optimisation, the reactive diluent content of resins produced initially was intentionally made lower than optimal ratios seen in literature<sup>25,56–58</sup>, this allowed a range of resins with varying amounts of reactive diluent to be tested by simply mixing in more of these monomeric reagents to the initially produced resins.

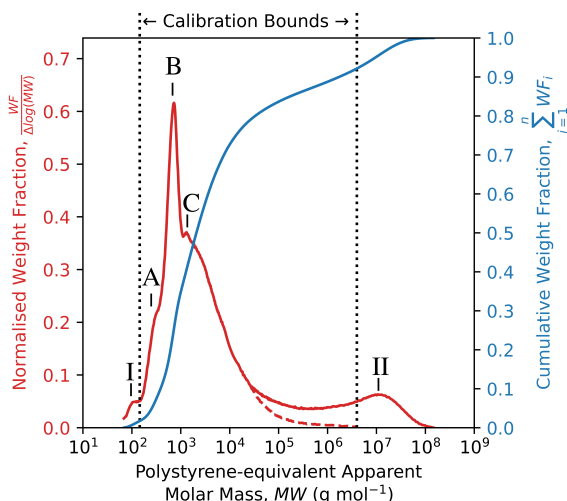
Conventional gel permeation chromatography (GPC) was used to characterise the distribution of hydrodynamic volumes in solution and to estimate polystyrene-equivalent apparent number- and weight-average molar masses ( $M_{n,app}$ ,  $M_{w,app}$ ) and dispersity ( $\bar{D} = M_w/M_n$ ). Testing was undertaken using a Viscotec system (manufactured by Malvern Panalytical) consisting of both a GPCmax autosampling module as well as the TDA305 triple-detection module (providing a concentration detector, viscometer, and light scattering detector). The column set employed consisted of a guard column followed by two Agilent PLgel 5  $\mu$ m MIXED-D analytical columns (all being held at 35 °C), with these providing high-molecular weight resolution typically necessary for the study of resins. Consistent

with previous GPC work on unsaturated polyesters we used tetrahydrofuran as the eluent, with this suitably solvating the unsaturated polyester samples obtained from the polyesterification reactor<sup>59</sup>. Polyester samples were diluted to 5 mg mL<sup>-1</sup> and filtered through an Agilent 0.45  $\mu$ m polytetrafluoroethylene membrane before entry into the GPC columns at an injection volume of 100  $\mu$ L. Five replicate injections were undertaken at a flow rate of 1 mL min<sup>-1</sup> with the data presented in Figure 4 representing one of these experiments.

Although viscometry and light-scattering signals were recorded, molar-mass distributions were calculated using the conventional refractive index based method with polystyrene calibration. This was necessitated by a combination of factors including the absence of a known refractive index increment value, the heterogeneity of the polyester samples handled (obfuscating signal responses across detectors), and the extremely broad elution range. Instead the viscometer and light-scattering traces were used qualitatively to assess the presence of large species. All molar masses reported are polystyrene-equivalent apparent values, as deduced using Agilent EasiVial PS-H calibration samples. This conventional calibration method has previously been shown to successfully approximate the number average molecular weight of polyesters, with values obtained from either vapour pressure osmometry or end group analysis being within 15% of GPC values<sup>60</sup>.



**Figure 3:** Simple reaction scheme for synthesis of poly(propylene itaconate).



**Figure 4:** A graph depicting the distribution of mass across various polymer chain lengths for the polyester sample tested using GPC. The dashed red line is an hypothetical reconstruction as presented in the supporting information’s section 9.4. I/II: Signals outside the calibrated range (total permeation/exclusion). A/B/C: Fitted components.

Figure 4 presents the results of the GPC characterisation of the batch of polyester synthesised, with both differential molecular weight fraction and cumulative weight fraction plotted against apparent molecular weight of chains. The apparent distribution is multi-modal with two peaks occurring outside of the calibration limits (Fig. 4I/II) and three peaks identifiable within (Fig. 4A/B/C).

Peaks I and II are caused by total permeation and exclusion respectively, with both being expected to be unphysical as they occur outside the of the styrene-standard calibration bounds, rendering molecular weights estimated here uncertain extrapolations. If an arbitrary set point of  $10^5 \text{ g mol}^{-1}$  is taken as the boundary separating the early eluting fraction from the rest of the trace, then approximately 15% of the cumulative weight

fraction is associated with the early-eluting feature at or near the column-exclusion limit. Its accompanying viscometry and light-scattering responses are consistent with species of unusually large hydrodynamic volume (in excess of the upper calibration bound,  $MW > 3.9 \times 10^6 \text{ g mol}^{-1}$ ), which may include highly branched species, soluble microgels or aggregates. Consequently, calculation of number  $M_n$  and weight  $M_w$  average molecular weights directly from the apparent distribution exhibit values ( $952$  &  $1.6 \times 10^6 \text{ g mol}^{-1}$ ) far in excess of that which might be expected for either conventional maleic-based ( $709$  &  $3,382 \text{ g mol}^{-1}$ ) or similar itaconate-based polyesters ( $600$  &  $3,657 \text{ g mol}^{-1}$ )<sup>38</sup>. A hypothetical reconstruction of the trace excluding the early-eluting feature (see the supporting information in Sec. 9.4) was plotted (red dashed line, Fig. 4). This hypothetical trace displayed a 15% reduction in its summed weight fraction relative to the apparent distribution and presented  $M_n$  and  $M_w$  values ( $606$  &  $3,006 \text{ g mol}^{-1}$ ) not dissimilar to those detailed in literature for similar itaconate based polyesters.

By discerning the existence of three separate peak structures (Fig. 4A/B/C), mathematical deconvolution was employed (see Sec. 9.4) to split apart the apparent distribution into three log normal distributions allowing the estimation of  $M_n$ ,  $M_w$ , weight fraction, and dispersity  $\mathbb{D}$  as summarised in Table 1. Distribution C represented the largest weight fraction of the modelled distributions, and exhibited a dispersity ( $\mathbb{D} = 5.62$ ) not dissimilar to that reported for similar itaconate-based polyester

| Distribution | $M_n \pm 1\sigma$<br>(g mol <sup>-1</sup> ) | $M_w \pm 1\sigma$<br>(g mol <sup>-1</sup> ) | $\mathbb{D} \pm 1\sigma$ |
|--------------|---------------------------------------------|---------------------------------------------|--------------------------|
| Averaged     | 673±19                                      | 971,273±16,834                              | 1,513±43                 |
| Apparent     | 952                                         | 1,559,183                                   | 1,638                    |
| Hypothetical | 606                                         | 3006                                        | 4                        |
| A            | 165                                         | 168                                         | 1                        |
| B            | 837                                         | 872                                         | 1                        |
| C            | 1,727                                       | 9,711                                       | 5                        |

Table 1: Summary of molecular weights and dispersity values for the average trace (averaged across the five injections), for the apparent trace analysed, the hypothetical reconstructed trace, and the deconvoluted (A/B/C) distributions obtained from the GPC of the polyester resin.

resins ( $\mathbb{D} = 6.0$ )<sup>38</sup>. Such dispersity values were likely the result of an extended reaction time (the outcome of eschewing catalysts) coupled with the lack of radical inhibiting agent, which would have facilitated a high degree of branching. Contrastingly, the modelled distributions A/B presented low dispersity values. These two peaks were consistent with a proportion of the linear polyester species undergoing intramolecular cyclisation, on the basis that cyclised species would present as a separate peak of lower molecular weight as their more compact conformation has a lower hydrodynamic volume relative to a linear species of the same molecular mass. Literature presents the occurrence of these peaks in GPC analyses for a variety of unsaturated polyester resins based on maleic and phthalic anhydride<sup>60–62</sup>, whilst the more recent work of Molina et al. further considers the possibility of substantial cyclisation in itaconate-based unsaturated polyesters<sup>63</sup>. Note that we could not rule out the contemporaneous elution of solvent or additive peaks within the trace.

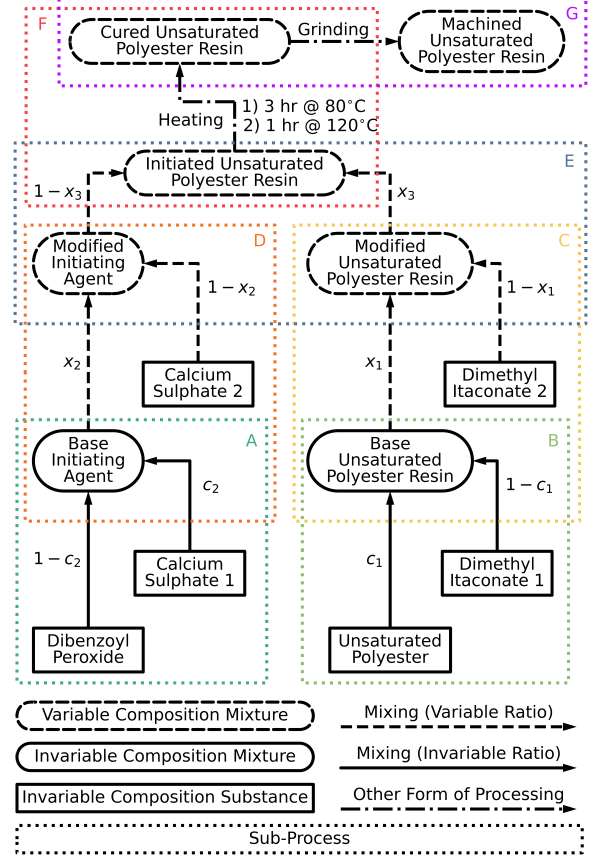
## 2.4 Composite Preparation

Figure 5 presents an overview of the method by which cured thermoset composite samples of various compositions were prepared. Given the danger of self-accelerating decomposition reactions and associated thermal

runaway, we obtained a safely filled base initiating agent comprised of both dibenzoyl peroxide and calcium sulphate (Fig. 5A); this arrived pre-mixed in a set ratio of 2:8 between the dibenzoyl peroxide ( $1 - c_2 = 0.2$  w/w) and the calcium sulphate ( $c_2 = 0.8$  w/w). Base unsaturated polyester resin was mixed (Fig. 5B) in a set ratio of 7:3 between unsaturated polyester resin ( $c_1 = 0.7$  w/w) and dimethyl itaconate ( $1 - c_1 = 0.3$  w/w) as described previously. This elected ratio represents the lowest amount of reactive diluent possible before increasing viscosity precluded the possibility of further mixing. This increased the size of the possible parameter space to be interrogated by increasing the range of modified unsaturated polyester resin reactive diluent concentrations that could be mixed to by means of dilution.

Modified unsaturated polyester resin was mixed (Fig. 5C) in a varying ratio of  $x_1 : (1 - x_1)$  between the base unsaturated polyester resin ( $x_1$  w/w) and the dimethyl itaconate ( $1 - x_1$  w/w). By varying the  $x_1$  parameter over the range  $([0.45, 1.00])$  we could produce any of the desired mixing ratios, fully describing the first predictor variable to be optimised ( $x_1$ , unsaturated polyester content modifier). Modified initiating agent was mixed (Fig. 5D) in a varying ratio of  $x_2 : (1 - x_2)$  between the base initiating agent ( $x_2$ ) and calcium

sulphate ( $1 - x_2$ ). The parameter  $x_2$  was varied over the range  $([0.4, 1.00])$  hence describing the second predictor variable to be optimised ( $x_2$ , dibenzoyl peroxide content modifier). Initiated unsaturated polyester resin was mixed (Fig. 5E) in a varying ratio of  $x_3 : (1 - x_3)$  between modified unsaturated polyester resin ( $x_3$ ) and modified initiating agent ( $1 - x_3$ );  $x_3$  was provided a range  $([0.5, 0.95])$  which therewith described the final predictor to be optimised ( $x_3$ , initiating agent content modifier).



**Figure 5:** *Mixing and processing schedule for the manufacture of thermoset composite samples. (A) Manufacturer’s mixing for base initiator. (B) In-house mixing for base resin. (C) Mixing for modified resin. (D) Mixing for modified initiator. (E) Mixing for initiated sample. (F) Thermochemical curing. (G) Machining.*

## 2.5 Composite Characterisation

Initiated unsaturated polyester resin samples were thermochemically cured into rigid thermoset plastic samples by exposing them to elevated temperatures in a drying oven (Fig. 5F); the curing profile was fixed for all samples: 3 hours at 80 °C followed by 1 h at 120 °C. All cured thermoset samples were machined to the same dimensions (Fig. 5G) in preparation for characterisation; this was achieved by a grinding process.

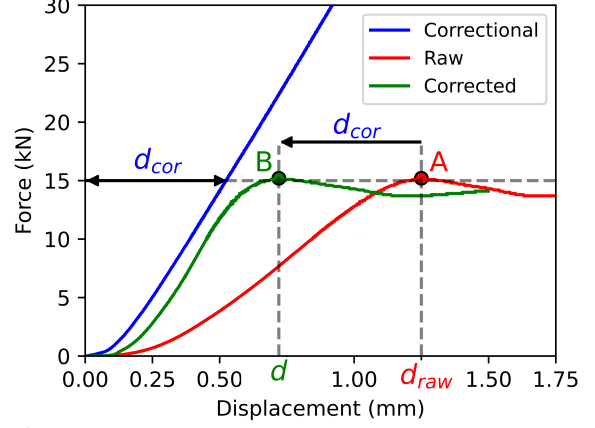
In order to quantify the mechanical performance of thermoset polymers created using Bayesian optimisation, their apparent compressive modulus ( $E_{app}$ ) and peak compressive stress ( $\sigma_{max}$ ) were characterised using compressive testing. Compressive testing was chosen based on both the simplicity with which a curing regime could be devised for the manufacture of samples of the requisite dimensions and the low mass of material required for carrying out tests. The operational metrics were chosen as stress-strain curves consistently failed to exhibit conventional initial linear regions or a distinct yield or break point. An American Society for Testing and Materials standard



was employed (“ASTM D695-23”) in the devising of method, with a cylindrical test sample diameter of 12.7 mm height 12.7 mm, and a compression speed of  $1.3 \text{ mm min}^{-1}$  adopted. To mitigate against concerns of Euler buckling taking place in samples of the proposed dimensions, we calculated the critical load based on Euler’s critical load equation (see Sec. 9.5 in the supporting information) and some estimated properties for conventional polyester thermosets<sup>64</sup>. What follows is a summary of the method undertaken in the automated analysis of datasets derived from mechanical testing, with more detail provided in the supporting information (see Sec. 9.5) .

**Equipment** Compressive testing was carried out using a “Tinius Olsen 50ST” universal test machine (50 kN load capacity) fitted with compression platens. Standard software (‘Horizon’) purveyed by the machine’s manufacturer was used in the control of the machine, retrieval of data from the load cells, and export of formatted data for further analysis. Exported data consisted of time ( $\pm 0.16 \text{ } \mu\text{s}$ ), force ( $\pm 1 \text{ mN}$ ), and platen displacement ( $\pm 1 \text{ nm}$ ).

**Correction of Raw Data** Finite compliance of the testing machine introduced an additional displacement contribution into raw exported datasets. To correct for this, a blank compression test was performed with no sample positioned between the platens, generating a force-displacement curve corresponding to the inherent compliance of the machine. This curve was subsequently used as a correctional dataset. Figure 6 compares the correctional dataset with the representative raw and corrected compression datasets. Point ‘A’ on the raw dataset is transformed to point ‘B’ on the corrected dataset by subtracting the corresponding machine displacement contribution  $d_{\text{cor}}$ , from the measured displacement  $d_{\text{raw}}$ .



**Figure 6:** Strain correction of an archetypical raw dataset; the legend notes the various datasets considered.

**Processing of Corrected Data** Load values from a sample’s corrected dataset were converted into engineering stress values by dividing by the initial surface area. Each cylindrical sample’s diameter was measured using a “Magnusson” digital caliper (accuracy  $\pm 0.02 \text{ mm}$ ) from three different positions ( $\phi_1, \phi_2, \phi_3$  in mm), each offset by roughly  $+45^\circ$  around the circumference, and the mean of these values ( $\phi_{\text{avg}}$  in mm) calculated. Subsequently, the average top-face surface area ( $A_{\text{avg}}$ ) was obtained from the average diameter  $\phi_{\text{avg}}$ , allowing the engineering stress ( $\sigma$  in MPa) to be calculated (Equation 1) by considering the force applied ( $F$  in N) and the initial top surface area,  $A_{\text{init}}$ <sup>65</sup>.

$$\sigma_{\text{eng}} = \frac{F}{A_{\text{init}}} \quad (1)$$

Displacement values from a sample’s corrected dataset were converted into engineering strain by taking the difference between the initial length of the samples and the length of the sample during compression. The initial length ( $l_{\text{init}}$  in mm) of sample was calculated by taking the mean of three length measurements ( $l_1, l_2, l_3$  in mm) (these taken in the same fashion as for diameter). Following this, by considering the length of a sample at any given point in time during a compressive test ( $l = l_{\text{init}} - d$ ), the engineering strain was calculated by finding the change in sample length ( $\Delta l_{\text{init}}$  in mm)

and dividing this by the initial length of the sample ( $l_{\text{init}}$ ) as shown in Equation 2.

$$\epsilon_{\text{eng}} = \frac{\Delta l_{\text{init}}}{l_{\text{init}}} = \frac{l - l_{\text{init}}}{l_{\text{init}}} \quad (2)$$

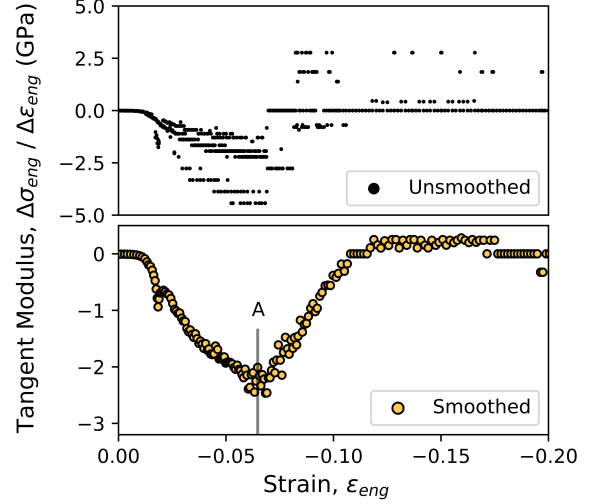
**Stress-Strain Smoothing** To produce a smoothed dataset (from which apparent compressive modulus and peak compressive stress could be deduced) a combination of sample-rate reduction and filtering were applied to downsample the stress-strain dataset. Downsampling was achieved using a windowed averaging technique, whereby the unsmoothed dataset was split into a number of windows along the strain-axis by electing a suitable number of points per window.

**Derivative Analysis** Because the stress-strain curves did not consistently exhibit a distinct initial linear region, a conventional Young’s modulus could not be determined robustly. An operational apparent compressive modulus,  $E_{\text{app}}$ , was therefore estimated from the maximum absolute local slope of the smoothed engineering stress-strain curve. The corresponding strain was located as the first inflection point of the stress-strain curve (equivalently, the minimum tangent modulus under the adopted compression sign convention). To find this inflection point, the first derivative of stress with respect to strain (i.e. tangent modulus  $E_t$  in MPa) was calculated from the smoothed stress-strain dataset by taking into account the instantaneous change in both stress ( $\Delta\sigma_{\text{eng}}$ ) and strain ( $\Delta\epsilon_{\text{eng}}$ ), and given by Equation 3.

$$E_t = \frac{\Delta\sigma_{\text{eng}}}{\Delta\epsilon_{\text{eng}}} = \frac{\sigma_{\text{eng}}^{j+1} - \sigma_{\text{eng}}^j}{\epsilon_{\text{eng}}^{j+1} - \epsilon_{\text{eng}}^j} \quad (3)$$

Figure 7 shows how smoothing of the stress-strain dataset facilitated the analysis of the resultant tangent modulus versus strain dataset. Step-like features are apparent in the tangent-modulus curve obtained from the unsmoothed stress-strain data. These may be consistent with either

distretisation or internal processing of the exported displacement signals and subsequently amplified by the process of numerical differentiation. To avoid undue physical significance being assigned to these artefacts, proportional-limit detection was performed upon a smoothed stress-strain dataset.

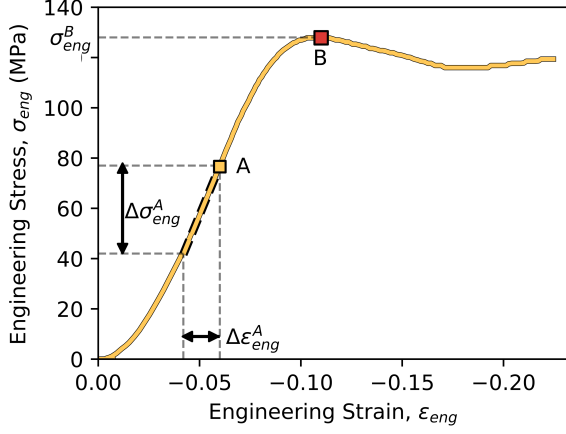


**Figure 7:** A comparison of the tangent modulus vs strain curves obtained from both the raw and the smoothed counterpart stress-strain datasets. (A) Strain at first inflection point.

Detection of the first inflection point was automated using an algorithm that iterated chronologically through the tangent modulus versus strain dataset from low to high strains. At each point, the algorithm calculates an average gradient exhibited by an equal number of points both before and after the current strain value (the ‘point horizon’). Using a point horizon of ten, the first inflection point was defined as the first point at which the summed forward and backward gradients fell below zero. This point was taken as the proportional limit, as shown in Figure 7A.

**Objective Retrieval** By re-considering the smoothed stress-strain dataset, the point with the closest proximity to the proportional limit’s determined strain value was equated to the first inflection point: the proportional limit (Fig. 8A).





**Figure 8:** Automatic determination of apparent compressive modulus and peak compressive stress. (A) First inflection point. (B) Second inflection point.

As shown in Figure 8, a linear regression was undertaken to identify the apparent compressive modulus  $E_{app}$  in accordance with Equation 4. The point nearest the identified strain was used to define the endpoint of a local fitting window. A linear regression was fitted to the ten preceding smoothed points, and  $E_{app}$  was taken as the absolute value of the fitted slope. Since the data were obtained in compression, the fitted gradients were negative; reported modulus values are given as absolute magnitudes.

$$E_{app} = |(\Delta\sigma_{eng}/\Delta\epsilon_{eng})_{fit}| \quad (4)$$

The peak compressive stress,  $\sigma_{max}$ , was defined as the maximum engineering stress recorded during the test:  $\sigma_{max} = \max(\sigma_{eng})$ . For these curves, the second-inflection-point routine located this maximum reproducibly, but the quantity should not be described as yield strength or the onset of plasticity.

For the stress-strain curves obtained in this study, this point coincided with the maximum engineering stress and could therefore be identified reproducibly across samples.

## 2.6 Optimisation

As described above, three predictors controlled the composition of samples: the unsaturated polyester content ratio ( $x_1$ ), the dibenzoyl peroxide content modifier ( $x_2$ ) and the initiating agent content modifier

( $x_3$ ). These three predictors and their ranges formed a  $d$ -dimensional ( $d = 3$ ) parameter space ( $\mathcal{X}$ ) which is a subset of the entire real coordinate space ( $\mathcal{X} \subseteq \mathbb{R}^d$ ). This parameter space was explored by the sequential optimisation techniques, with the objective variables including apparent compressive modulus ( $E_{app}$ ) and peak compressive stress ( $\sigma_{max}$ ).

Three different optimisation procedures were carried out from the same initial prior dataset: two were single-objective Bayesian routines with differing objectives, whilst the third was a purely exploratory method which was objective-indifferent. A fourth multi-objective Bayesian optimisation routine incorporating both the objectives previously considered was carried out using a combination of the original prior dataset and all data generated by the three sequential optimisations.

**Single-Objective Experiments** Optimisation problems were based on black-box functions  $f$  which took any point within the parameter space  $\mathcal{X}$  and returned a real-valued target  $\mathbb{R}$ ; this being a mapping between the two respective spaces ( $f : \mathcal{X} \rightarrow \mathbb{R}$ ). Equation 5 describes optimisation as the maximisation of the black-box function by tuning the predictor variable values ( $\mathbf{x} = [x_1, x_2, x_3]$ ) such that the global optimum set of predictors ( $\mathbf{x}^* = [x_1^*, x_2^*, x_3^*]$ ) is sought. The finite experimental campaign cannot establish the global maximiser; subsequent results are therefore described in terms of the best observed formulation within the sampled design space and sample budget.

$$\mathbf{x}^* = \arg \max_{\mathbf{x} \in \mathcal{X}} f([x_1, x_2, x_3]) \quad (5)$$

Equation 6 describes the two black-box functions which were separately maximised during the course of the single-objective optimisations; the former  $f_E$  and latter  $f_\sigma$  considering apparent compressive modulus  $E_{app}$  and peak compressive stress  $\sigma_{max}$  as target variables respectively.

$$\begin{aligned} E_{\text{app}} &= f_E(\mathbf{x}) \\ \sigma_{\text{max}} &= f_\sigma(\mathbf{x}) \end{aligned} \quad (6)$$

Eight vectors of predictor variables, each corresponding to a sample ( $i$ ), were representatively drawn by Sobol sampling from across the parameter space  $\mathcal{X}$  forming an initial matrix of predictors ( $\mathbf{X}$ ); the matrix's form is presented in Equation 7.

$$\begin{aligned} \mathbf{X} &= [\mathbf{x}^{i=1}, \mathbf{x}^{i=2}, \dots, \mathbf{x}^{i=8}] \\ &= [[x_1^{i=1}, x_2^{i=1}, x_3^{i=1}], [x_1^{i=2}, \dots], \dots] \end{aligned} \quad (7)$$

Evaluation of all the vectors of the predictor matrix ( $f_E(\mathbf{X})$  or  $f_\sigma(\mathbf{X})$ ) generated a vector of targets ( $\mathbf{E}_{\text{app}}$  or  $\sigma_{\text{max}}$ ). Combination of the initial predictor matrix  $\mathbf{X}$  and the corresponding targets  $f(\mathbf{X})$  produced an initial dataset ( $\mathcal{D} = [\mathbf{X}, f(\mathbf{X})]$ ) with which sequential optimisation experiments were begun. During the course of a sequential optimisation experiment, 16 further samples were drawn from across the parameter space  $\mathcal{X}$ , with one sample taken per iteration; each sample's predictor vector  $\mathbf{x}$  and function evaluation  $f(\mathbf{x})$  was appended to the growing dataset  $\mathcal{D}$  associated with that experiment. At each iteration, the Bayesian experiments strove to balance exploration and exploitation during selection of sample composition within the parameter space, whilst the benchmark deterministic experiment solely focussed upon exploration.

Bayesian optimisation involved an iterative process of developing a surrogate model approximating the latent objective function based on the available dataset  $\mathcal{D}$ , using an acquisition function to evaluate candidate suitability and selection of a new sample  $\mathbf{x}^i$ , and finally evaluation of a sample followed by incorporation of the resultant target scalar  $\sigma_{\text{max}}^i$  and predictor vector  $\mathbf{x}^i$  into the growing dataset  $\mathcal{D}$ .

On updating the dataset  $\mathcal{D}$ , Gaussian process regression was used to develop a Gaussian process regression model  $\mathcal{GP}$  which

can be considered a distribution over possible functions ( $f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), K(\mathbf{x}, \mathbf{x}'))$ ) fully described by both its mean ( $m$ ) and covariance ( $K$ ) functions. Computationally, the Gaussian process was modelled by a high-dimensional multivariate Gaussian distribution  $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  described by its mean vector  $\boldsymbol{\mu}$  and covariance matrix  $\boldsymbol{\Sigma}$ ; each successive iteration saw the prior process conditioned on new data producing an updated posterior process.

To encode our prior beliefs in process smoothness, we employed the Matérn family of covariance functions; specifically, one with a hyperparameter  $\nu = 5/2$ . The  $\nu = 5/2$  case balanced the need for smoothness in sampled paths with the drawbacks of using higher values of  $\nu$ . As  $\nu \rightarrow \infty$ , the sampled paths become infinitely differentiable, making them unrealistically smooth for practical application<sup>66</sup> and potentially interfering with successful calculation of the likelihood<sup>67</sup>.

$$K_{M^{5/2}}(\mathbf{x}, \mathbf{x}') = (1 + \sqrt{5}d + \frac{5}{3}d^2)\exp(-\sqrt{5}d) \quad (8)$$

where  $d = |\mathbf{x} - \mathbf{x}'|$

Each iteration saw the use of a one-step lookahead sampling policy utilising a simple reward utility function (Equation 9) which takes the current dataset  $\mathcal{D}$  and of all samples  $\mathbf{X}$  returns the sample  $\mathbf{x}$  exhibiting the highest target ( $\phi$ ; in practice either  $E_{\text{app}}$  or  $\sigma_{\text{max}}$ ) according to the posterior process's mean function  $\max m(X)$ .

$$u(\mathcal{D}) = \max m(\mathbf{X}) \quad (9)$$

Utility quantification of a putative sample  $\mathbf{x}_p$  was calculated by the expected improvement acquisition function  $\alpha_{\text{EI}}$  (Equation 10) which calculates the marginal gain in utility ( $u(\mathcal{D}') - u(\mathcal{D})$ ) between the incumbent best target value in the current dataset ( $\phi^*$ ) and that of a new dataset containing the putative  $\mathbf{x}_p$  value ( $u(\mathcal{D}')$ ). The second line of Equation 10 clarifies how the calculation is one for the expected marginal gain, whilst the third presents reformulation for the noiseless case; this equation exemplifies how putative points are rewarded based on both scale and probability of improvement

effectively trading off exploitation and exploration. Here, we applied a more sophisticated logarithmic noise-aware implementation (“qLogNoisyExpectedImprovement” or “qLNEI”) which has recently seen successful use in high-dimensional computational domains<sup>68</sup>. Given a lack of repeat measurements during characterisation, target variable values provided to the Ax package were modelled with inferred homoscedastic noise.

$$\begin{aligned}\alpha_{\text{EI}}(\mathbf{x}_p, \mathcal{D}) &= u(\mathcal{D}') - u(\mathcal{D}) \\ &= \mathbb{E}[\max m_{\mathcal{D}'}(\mathbf{X}')] - \max m_{\mathcal{D}}(\mathbf{X}) \\ &= \int_{\phi^*}^{\infty} \phi \mathcal{N}(\mu, \sigma^2) d\phi - \phi^* \int_{\phi^*}^{\infty} \mathcal{N}(\mu, \sigma^2) d\phi\end{aligned}\quad (10)$$

Trivial optimisation of the acquisition function (Equation 11) was achieved by generating a large set of Sobol sampled points across the parameter space  $\mathbf{X}_p$ , after which cheap evaluation facilitated identification of the candidate sample with the maximal utility score  $\mathbf{x}^+$ .

$$\mathbf{x}^+ = \arg \max_{\mathbf{x}_p \in \mathcal{X}} \alpha(\mathbf{X}_p) \quad (11)$$

The objective-indifferent space-filling benchmark sampling exercise was based on a sequential Monte Carlo strategy by the name of mc-intersite-proj-th (“MIPT”)<sup>69</sup>. Utility was quantified by the conditional acquisition function displayed in Equation 12, which took both a putative point  $\mathbf{x}_p$  and the current dataset’s previous sample coordinates  $\mathbf{X}$  and returns either zero or the Euclidean distance to the nearest previously sampled point  $k$  depending on whether projected distance is below or in excess of the  $d_m$  value respectively. The  $d_m$  is dependent upon the number of previous samples  $n$  and the arbitrary constant  $s$  which has been found to be optimally set to 0.5 for a reasonable trade-off between intersite distance and projected distance.

$$\begin{aligned}\alpha_{\text{mipt}}(\mathbf{x}_p, \mathbf{X}) &= \\ \begin{cases} 0 & \text{if } \min_{\mathbf{x}_i \in \mathbf{X}} \|\mathbf{x}_i - \mathbf{x}_p\|_{-\infty} < d_m \\ k, & \text{if } \min_{\mathbf{x}_i \in \mathbf{X}} \|\mathbf{x}_i - \mathbf{x}_p\|_{-\infty} \geq d_m \end{cases} \quad (12) \\ \text{where } d_m &= \frac{2s}{n} \text{ and } k = \min_{\mathbf{x}_i \in \mathbf{X}} \|\mathbf{x}_i - \mathbf{x}_p\|_2\end{aligned}$$

**Multi-Objective Experiment** Given the desirability of both apparent compressive modulus and peak compressive stress in a thermoset composite, a dual-objective Bayesian optimisation experiment was attempted, with the *a posteriori* goal of sample-efficiently identifying the Pareto front ( $\mathcal{P}$ )<sup>70</sup>. The Pareto front consists of all samples which remain unsurpassed (by any other sample evaluated) in terms of at least one of their target variables ( $E_{\text{app}}$  or  $\sigma_{\text{max}}$ ). This Pareto front allows a rational agent to make subjective yet informed decisions about material composition in light of expected trade-offs in end properties. Equation 13 frames the multi-objective optimisation problem in terms of a dual-objective function  $\mathbf{f}$  (comprised of the single-objective functions  $f_E$  and  $f_\sigma$ ) which returns a vector  $\mathbf{y}$  of the two targets ( $E_{\text{app}}$  and  $\sigma_{\text{max}}$ ).

$$\begin{aligned}\mathbf{y} &= \mathbf{f}(\mathbf{x}) \\ \{E_{\text{app}}, \sigma_{\text{max}}\} &= \{f_E(\mathbf{x}), f_\sigma(\mathbf{x})\}\end{aligned}\quad (13)$$

An observed Pareto front may be obtained directly from a given dataset  $\mathcal{D}$  using the function  $f_{\mathcal{P}}$  (Eq. 14).

$$\begin{aligned}\mathcal{P} &= f_{\mathcal{P}}(\mathcal{D}) \\ &= \{\mathbf{y} | \mathbf{y} \in \mathbf{Y}, \nexists \mathbf{y}' \in \mathbf{Y} \text{ s.t. } \mathbf{y}' \succ \mathbf{y}\}\end{aligned}\quad (14)$$

Alternatively, where a multi-objective function  $\mathbf{f}$  is considered alongside a matrix of predictor variables  $\mathbf{X}$ , the function  $f_{\mathcal{P}}^*$  may be used to produce a Pareto front (Eq. 15).

$$\begin{aligned}\mathcal{P} &= f_{\mathcal{P}}^*(\mathbf{X}; \mathbf{f}) \\ &= \{\mathbf{f}(\mathbf{x}) | \mathbf{x} \in \mathbf{X}, \mathbf{f}(\mathbf{x}) \succ \mathbf{f}(\mathbf{x}') \forall \mathbf{x}' \in \mathbf{X}\}\end{aligned}\quad (15)$$

By retrieving all the raw mechanical test data associated with the samples evaluated in the three sequential experiments and their common prior, a 56-sample dataset  $\mathcal{D}$  was generated comprised of a matrix of sampled predictors  $\mathbf{X}$  and a matrix of their respective targets  $\mathbf{Y}$  ( $E_{\text{app}}$  and  $\sigma_{\text{max}}$ ). Having conditioned two separate Gaussian processes on  $E_{\text{app}}$  and  $\sigma_{\text{max}}$  we obtained two separate posterior processes ( $\mathcal{GP}_E$  and  $\mathcal{GP}_\sigma$ ); these provided initial prior processes for all subsequent sequential optimisation. The pair of resulting processes were respectively described mathematically by their mean ( $m_E$ ,  $m_\sigma$ ) and covariance functions ( $K_E$ ,  $K_\sigma$ ) and

computationally by their multivariate normal analogues ( $\boldsymbol{\mu}_E$ ,  $\boldsymbol{\mu}_\sigma$  and  $\boldsymbol{\Sigma}_E$ ,  $\boldsymbol{\Sigma}_\sigma$ ). Equation 16 shows how a dual-objective surrogate function  $\tilde{\mathbf{f}}$  may be sampled from the pair of Gaussian processes ( $\mathcal{GP}_E$  and  $\mathcal{GP}_\sigma$ ) previously conditioned.

$$\begin{aligned} \tilde{\mathbf{f}} &\sim \{\mathcal{GP}_E, \mathcal{GP}_\sigma\} \\ \{\tilde{f}_E, \tilde{f}_\sigma\} &\sim \{\mathcal{GP}(m_E, K_E), \mathcal{GP}(m_\sigma, K_\sigma)\} \end{aligned} \quad (16)$$

The utility of a given Pareto front was evaluated using the hypervolume indicator utility function  $u_{\text{hv}}$  presented in Equation 17, which is equal to the  $d$ -dimensional Lebesgue measure  $\lambda_d$  of the space dominated by a Pareto front  $\mathcal{P}$  and bounded from below by reference point  $r$  ( $r$  and  $v$  describe vertices).

$$u_{\text{hv}}(\mathcal{P}; r) = \lambda_d\left(\bigcup_{v \in \mathcal{P}} [r, v]\right) \quad (17)$$

Equation 18 shows how the utility function  $u_{\text{hv}}$  is used in the subsequent calculation of the one-step marginal gain known as the “hypervolume improvement” between the extant  $\mathcal{P}$  and the updated Pareto front  $\mathcal{P}'$  (which includes the putative point  $\mathbf{x}_p$ ).

$$f_{\text{hvi}}(\mathcal{P}, \mathcal{P}'; r) = u_{\text{hv}}(\mathcal{P}'; r) - u_{\text{hv}}(\mathcal{P}; r) \quad (18)$$

Here, we use a common multi-objective acquisition function called expected hypervolume improvement, described by Equation 19, which takes a putative point  $\mathbf{x}_p$  and the current dataset  $\mathcal{D}$  and returns the instantaneous improvement metric. Expectation manifests itself during the speculative update of the dataset  $\mathcal{D}$  to  $\mathcal{D}'$ , where the two targets are drawn from their respective marginal Gaussian distributions (of their respectively updated posterior Gaussian processes) located at the putative point  $\mathbf{x}_p$  whose expectation ( $\mathbb{E}(\mathbf{x}_p)$ ) is the mean of the marginal at that location ( $m_E(\mathbf{x}_p)$  or  $m_\sigma(\mathbf{x}_p)$ ).

$$\begin{aligned} \alpha_{\text{ehvi}}(\mathbf{x}_p; \mathcal{D}) &= f_{\text{hvi}}(f_{\mathcal{P}}(\mathcal{D}'), f_{\mathcal{P}}(\mathcal{D}); r) \\ \text{where } \mathcal{D}' &= \mathcal{D} \cup \{\mathbf{x}_p, m_E(\mathbf{x}_p), m_\sigma(\mathbf{x}_p)\} \end{aligned} \quad (19)$$

This work adopted Daulton’s reformulation of the classic acquisition function (Eq. 19) which takes into account the

noisy nature of objective function evaluations<sup>71</sup>. Although the true formulation contains an integral which is analytically intractable, it may be approximated using Monte Carlo integration. Equation 20 describes the integral-approximated reformulation of the acquisition function  $\hat{\alpha}$  which takes a putative point  $\mathbf{x}_p$  and the dataset  $\mathcal{D}$  and returns an instantaneous improvement value. This noisy expected hypervolume improvement acquisition function iterates from  $1, \dots, N$  using the  $t$  counter, with each iteration drawing a random dual-objective function  $\tilde{\mathbf{f}}_t$  from the posterior gaussian process (see Eq. 16). This function  $\tilde{\mathbf{f}}_t$  is used to speculatively produce Pareto fronts ( $\mathcal{P}$  and  $\mathcal{P}'$ ) for both an original dataset  $\mathbf{X}$  and the updated dataset  $\mathbf{X} \cup \mathbf{x}_p$  respectively, before the hypervolume improvement is gauged using  $f_{\text{hvi}}$  and the many trials  $N$  averaged.

$$\begin{aligned} \hat{\alpha}_{\text{nehvi}}(\mathbf{x}_p; \mathbf{X}) &= \\ \frac{1}{N} \sum_{t=1}^N f_{\text{hvi}}(f_{\mathcal{P}}^*(\mathbf{X} \cup \mathbf{x}_p; \tilde{\mathbf{f}}_t), f_{\mathcal{P}}^*(\mathbf{X}; \tilde{\mathbf{f}}_t); r) \end{aligned} \quad (20)$$

This work practically applied the log implementation of the noise-aware  $\alpha_{\text{nehvi}}$  acquisition function `qLogNoisyExpectedHypervolumeImprovement` (“qLNEHVI”) which has been shown to improve Bayesian optimisation performance in noisy settings<sup>71</sup>.

**Implementation** All Bayesian experiments were implemented in the Python3 programming language, with functionality spanning initial sampling techniques, adaptive sequential sampling methods, and an experimental workflow management system facilitating human-in-the-loop laboratory work; this package is given the acronym “BO4ACST” (version 1.0.0)- “Bayesian optimisation for atmospheric carbon sequestering thermosets”<sup>72</sup>. Note that Bayesian sequential optimisation techniques including Gaussian process regression, gaussian process conditioning, and sample utility evaluation were facilitated by the open-source adaptive experimentation platform or “Ax” (version 1.1.2)<sup>73</sup>. Note that “Ax” makes

use of a great deal of code from “Botorch”, especially in its implementation of custom Bayesian optimisation routines (e.g. non-standard acquisition function implementations)<sup>74</sup>.

### 3 Results

All results stem from the datasets generated during the practical-application of Bayesian optimisation in the domain of thermoset composite property development. Practical experimental results were split into those derived from the single-objective experiments and the singular multi-objective optimisation.

#### 3.1 Single-Objective Experiments

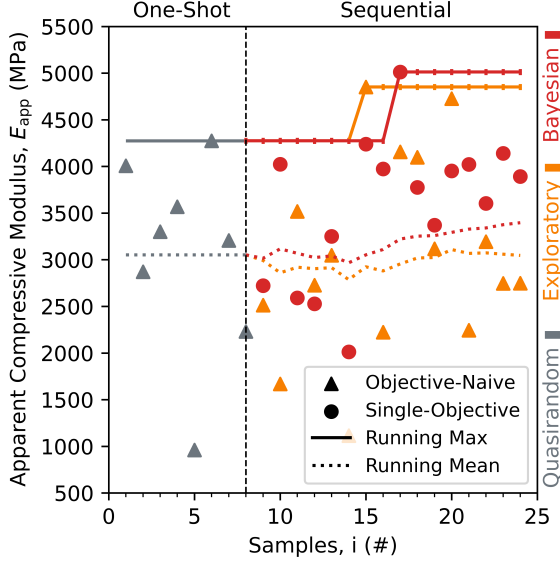
Two single-objective Bayesian optimisation experiments were performed, each using a different objective variable for sequential sampling: apparent compressive modulus  $E_{\text{app}}$  or peak compressive stress  $\sigma_{\text{max}}$ . Initial Gaussian process priors developed for both experiments were trained on the same eight quasirandomly sampled (Sobol) points  $\mathbf{X}$  (from across the parameter space  $\mathcal{X}$ ) and their evaluations  $\mathbf{Y}$ . Both experiments proceeded sequentially with a single sample evaluated at each Bayesian iteration. The sample limit was set to 24 (including the prior-forming dataset), and the optimisation was one of objective metric maximisation.

The objective-naive MIPT sampling technique was used to sample in a purely exploratory fashion, which provided a benchmark against which Bayesian optimisation routine progression could be gauged. Given its objective-naivety, purely exploratory nature, and the fact that it was initiated on the same prior-forming dataset as the Bayesian routines, the MIPT routine was only run once (samples having both apparent compressive modulus and peak compressive stress evaluated). The resulting dataset

could be directly compared with both the single- and multi-objective Bayesian sequential routines (see Section 3.2).

#### Apparent Compressive Modulus Experiment

Figure 9 compares the single-objective optimisation progression of both Bayesian and exploratory sequential routines in the maximisation of apparent compressive modulus. The one-shot quasirandomly-sampled initial dataset provided a baseline maximum of 4,274 MPa and a mean of 3,052 MPa. On reaching the sample limit, the single-objective Bayesian technique had exceeded both these baselines (max 5,012 MPa, mean 3,396 MPa) whilst the exploratory sampler only exceeded the maximum baseline (max 4,851 MPa, mean 3,046 MPa). Although the single-objective Bayesian routine ultimately discovered a sample exhibiting the highest apparent compressive modulus within this experiment, it is interesting that the exploratory routine discovered a greater number of points surpassing the baseline maximum set by the initial dataset. The exploratory sampling routine was the first to locate a sample improving upon the baseline maximum: sample 15 with an apparent compressive modulus of 4,851 MPa. Subsequently, at sample 17, the Bayesian single-objective routine discovered the experiment’s highest apparent compressive modulus of 5,012 MPa.

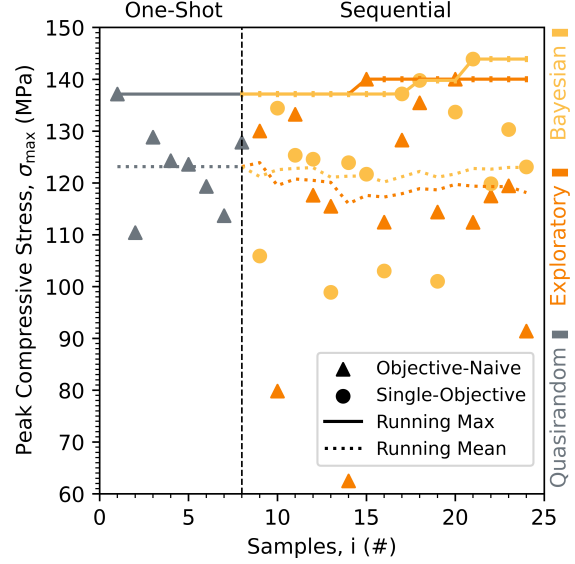


**Figure 9:** Progression of the sequential optimisation experiment concerning the maximisation of apparent compressive modulus. Initial one-shot samples drawn quasirandomly are presented separately from the two sequential techniques which they prime (by providing the initial datasets on which they are predicated). These two sequential techniques are either objective-naive (MIPT) or single-objective (Bayesian).

### Peak Compressive Stress Experiment

Figure 10 compares optimisation progression of both the Bayesian single-objective and exploratory sequential routines with respect to the maximisation of the peak compressive stress. The one-shot quasirandomly-sampled initial dataset provided a baseline maximum peak compressive stress of 137 MPa and a mean of 123 MPa. On reaching the sample limit, the Bayesian technique had surpassed the baseline maximum (143 MPa) and remained level with respect to the mean (123 MPa), whilst the exploratory routine had surpassed the baseline maximum (140 MPa) and failed to improve upon the baseline mean (118 MPa). In this experiment both the Bayesian single-objective and exploratory routines discovered the same number of points surpassing the baseline maximum, although the Bayesian technique ultimately uncovered the sample with the highest peak compressive stress. The exploratory sampling technique was the first to discover a point improv-

ing upon the maximum baseline; with this happening on sample 15, with a peak compressive stress of 140 MPa. Subsequently, the Bayesian sequential technique improves upon the maximum baseline on sample 18 (140 MPa) and surpasses the exploratory routine on sample 21, with a final peak compressive stress of 143 MPa.



**Figure 10:** Progression of a sequential optimisation experiment concerning the maximisation of peak compressive stress. Initial one-shot samples drawn quasirandomly are presented separately from the two sequential techniques which they prime (by providing the initial datasets on which they are predicated). These two sequential techniques are either objective-naive (MIPT) or single-objective (Bayesian).

### Comparison of Bayesian and MIPT Techniques

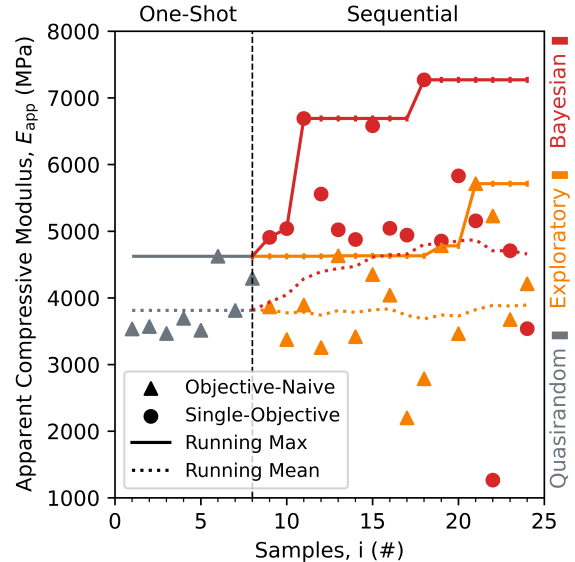
In this single experimental realisation, the Bayesian routines identified the highest observed value for each objective within the 24-sample budget, whereas MIPT identified near-best formulations earlier. Bayesian techniques benefit from adaptive sampling based on the balancing of both exploitative and exploratory behaviour, the former exhibited clearly in the non-decreasing trends shown by the Bayesian running means seen in both Figures 9&10. The margins over MIPT were modest: 3.3% for  $E_{app}$  (5,012 versus 4851 MPa) and 2.1% for  $\sigma_{max}$  (143 versus 140 MPa). Independently replicated manufacture/testing or re-

peated optimisation trajectories would be required to describe these differences as statistically significant.

In these trajectories, MIPT exceeded the initial baseline earlier than the Bayesian routines: this exploratory technique improved upon the baseline maxima ( $E_{\text{app}} = 4274$  MPa and  $\sigma_{\text{max}} = 137$  MPa) of both experiments on drawing sample 15 for evaluation ( $E_{\text{app}} = 4851$  and  $\sigma_{\text{max}} = 140$ ). The fact that this occurred on the 15<sup>th</sup> sample for both experiments is trivially explained by the fact that this was the same exploratory routine being performed. As opposed to the two single-objective routines seen in Figure 9 and Figure 10 which were sequentially drawing unique sample in attempts to maximise different properties, the objective-naive MIPT sampler was only run once with each sequential sample characterised in terms of both target variables. This is why the especially well-performing sample 15 appears in both progression plots (Figs. 9/10). A hypothesis that the exploratory sampler more efficiently improves upon the baseline maximum than the Bayesian routine would be weakly supported by the above findings given that both circumstances were precipitated from the same data point, but further repeat studies in the physical or computational domain would aid in the supporting or negating this conjecture.

**Styrenated Apparent Compressive Modulus Experiment** An experiment identical to that of the single-objective Bayesian optimisation for apparent compressive modulus was carried out with the conventional reactive diluent styrene in place of the dimethyl itaconate. Figure 11 shows an optimisation progression plot for the styrene-based system. The single-objective Bayesian optimisation technique shows strong performance in terms of both sample efficiency and scale of improvement above the baseline. On conclusion of this sequential optimisation, the best observed styrene-diluted formulation (sample 18) exhibited an apparent

compressive modulus of 7,271 MPa. Interestingly, the corresponding bio-sourceable-resin formulation experiment produced a sample (sample 17 in Fig. 9) exhibiting 69% the apparent compressive modulus of the best observed styrene-diluted formulation.



**Figure 11:** Progression of a sequential optimisation experiment concerning the maximisation of apparent compressive modulus with a styrene-based thermoset.

### 3.2 Multi-Objective Experiment

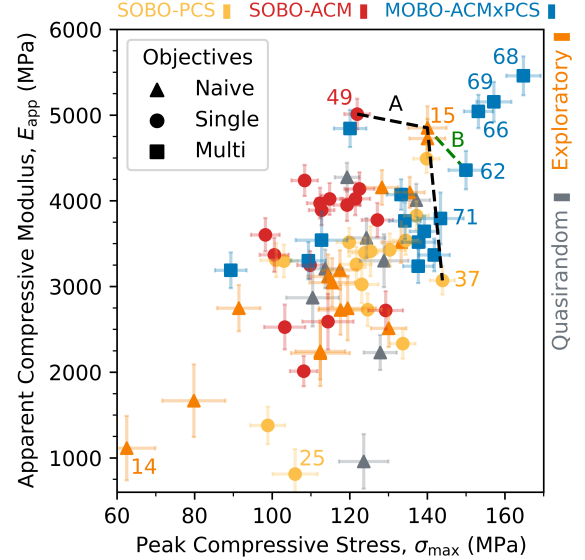
In the multi-objective setting, a Bayesian sequential routine was performed which made use of the noisy expected hypervolume improvement acquisition function to sample-efficiently identify the Pareto front within an objective space comprised of both apparent compressive modulus and peak compressive stress. This dual-objective optimisation was facilitated by two separate Gaussian processes modelling the two different objective variables. Both were trained on 56 samples concatenated from the re-analysed raw data of the single-objective experiments described in Section 3.1. The order of concatenation (and re-indexing of sample numbers) was as follows: one-shot objective-naive quasirandom sampler (samples 1-8), sequential exploratory objective-naive routine (9-24), sequential Bayesian peak compressive stress maximising routine (25-40), and the



sequential Bayesian apparent compressive modulus maximising routine (41-56). This prior-forming dataset  $\mathcal{D}$  yielded an initial Pareto front  $\mathcal{P}$  comprised of three samples: two having been discovered as maxima by Bayesian sequential routines in their two respective single-objective experiments with peak compressive stress (sample 37) and apparent compressive modulus (sample 49) as their objectives. Whilst the third (sample 15) was derived from the exploratory technique as a point which performed well in both objectives of interest. With an ultimate sample limit of 71 and a single sample drawn per Bayesian iteration, the sequential multi-objective Bayesian routine proceeded to develop the Pareto front over 15 further samples (57-71).

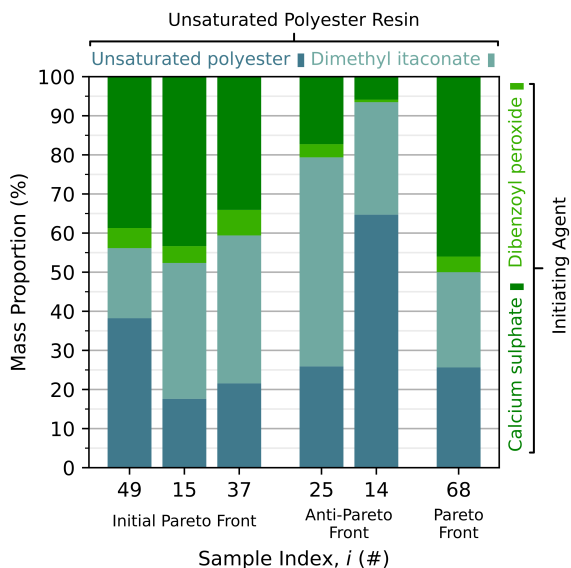
All samples both prior-forming and sequentially-obtained are depicted in Figure 12, with the initial Pareto front marked as a dashed black line (Fig.12A). Given the initial Pareto front and considering only the direct evaluations obtained for samples it can be said that five samples were discovered which improved upon the baseline described by the Pareto front (samples 62,66,68,69,71). Under progression of the multi-objective routine, the first novel pareto optimal point which improved the Pareto front was the 62<sup>nd</sup> sample which dominated the 37<sup>th</sup> sample shifting the boundaries of the developing Pareto front (Fig.12B). Subsequently, the sample 66 evaluated ( $E_{app} = 5,043$  MPa and  $\sigma_{max} = 153$  MPa) presented a marginal improvement upon both the apparent compressive modulus of sample 49 ( $E_{app} = 5,012$  MPa) and the peak compressive stress of sample 62 ( $\sigma_{max} = 150$  MPa), collapsing the Pareto front to a single Pareto optimal point. At culmination, sample 68 was the single non-dominated formulation among the samples evaluated (exhibiting  $E_{app} = 5,458$  MPa and  $\sigma_{max} = 165$  MPa), marking a simultaneous improvement upon the previous Pareto front. This final result contrasts sharply with the anti-Pareto front comprised of the 14<sup>th</sup> ( $E_{app} = 1,114$  MPa

and  $\sigma_{max} = 62$  MPa) and 25<sup>th</sup> ( $E_{app} = 813$  MPa and  $\sigma_{max} = 105$  MPa) samples. How the acquisition function likely navigated the parameter space and came to the samples it discovered has been reviewed from the perspective of the final Gaussian processes developed and is appended in the supporting information.



**Figure 12:** Results from a multi-objective Bayesian optimisation routine. Error bars correspond to the 68% posterior confidence intervals as sampled from the posterior Gaussian processes on conclusion of the multi-objective routine. (A) Initially observed Pareto front as a dashed black line. (B) Observed Pareto front improvement as a dashed green line. PCS, Peak Compressive Stress. ACM, Apparent Compressive modulus. SOBO, Single-objective Bayesian optimisation. MOBO, Multi-objective Bayesian optimisation.

Figure 13 presents the strong relation between initiator/filler-premix fraction maximisation and improvements to sample optimality: the average content is 13% for the anti-Pareto front, 44% for the initial Pareto front, and 50% for the final non-dominated formulation (sample 68). Furthermore, the initiator/filler-premix fraction also described a somewhat weaker relation between dibenzoyl peroxide marginalisation and sample optimality: anti-Pareto front 13.3%, initial Pareto front 12.3%, and 8% for the final non-dominated formulation. No analogously clear trend existed with respect to the unsaturated polyester resin composition.



**Figure 13:** Compositional breakdown for samples of interest including the initial Pareto front, the final anti-Pareto front, and the final Pareto front.

## 4 Conclusion

This work shows that particulate composites based on a fully bio-sourcable thermoset resin can be successfully optimised with respect to their apparent compressive modulus and peak compressive stress by adapting their composition during manufacture. The multi-objective campaign employing the log noisy expected hypervolume improvement acquisition function identified a composition that simultaneously improved  $E_{app}$  and  $\sigma_{max}$ , yielding the single non-dominated formulation among the 71 samples evaluated (sample 68 exhibiting  $E_{app} = 5,458$  MPa and  $\sigma_{max} = 165$  MPa).

In the domain of a sample-constrained single-objective Bayesian optimisation, using the log-noisy expected improvement acquisition function, a composite based on the fully bio-sourcable resin was obtained which exhibited 69% of the apparent compressive modulus of a conventional styrene-based material which had been optimised in the same fashion. Within matched 24-sample campaigns, the best observed dimethyl-itaconate formulation retained 69% of the apparent compressive modulus of the best observed styrene-diluted analogue. The multi-

objective formulation retained 75% of that comparator’s apparent modulus. Although there exist no directly comparable studies concerning these differences, the extent to which this bio-sourcable reactive diluent has managed to partially substitute this conventional fossil-derived monomer benchmark is notable.

The present study optimised compressive properties only; tensile, flexural, fracture, thermal, and ageing properties remain to be assessed. Replicate testing at selected formulations would further quantify experimental noise and improve  $\mathcal{GP}$  noise estimation. Finally, although the resin chemistry is bio-sourcable, full sustainability assessment would require lifecycle analysis of monomer production, curing, filler sourcing, and end-of-life processing. Given the interesting functional relationships witnessed within the parameter space with respect to predictor  $x_1$ , which concerns the balance between unsaturated polyester and reactive diluent, future work could examine how reactive diluents impact the mechanical properties of these materials. An optimisation of the styrene-based analogues in the single-objective setting with respect to the objective of peak compressive stress would also be a worthwhile study.

## 5 Data availability

Custom software developed for the purpose of execution of optimisation routines is available under the ‘BO4ACST’ project lodged in github at the following address: [https://github.com/ThomasDodd97/BO4ACST\\_Ax1](https://github.com/ThomasDodd97/BO4ACST_Ax1).

The latex project associated with this paper has been made available under the ‘UoC\_Paper2’ project lodged in github at the following address: [https://github.com/ThomasDodd97/UoC\\_Paper2](https://github.com/ThomasDodd97/UoC_Paper2).

Furthermore, it may be noted that the ‘Sandbox’ directory within the ‘UoC\_Paper2’ project contains both the data and the jupyter notebooks required to process it into

the graphics depicted in the present work.

## 6 Conflicts of Interest

There are no conflicts to declare.

## 7 Acknowledgements

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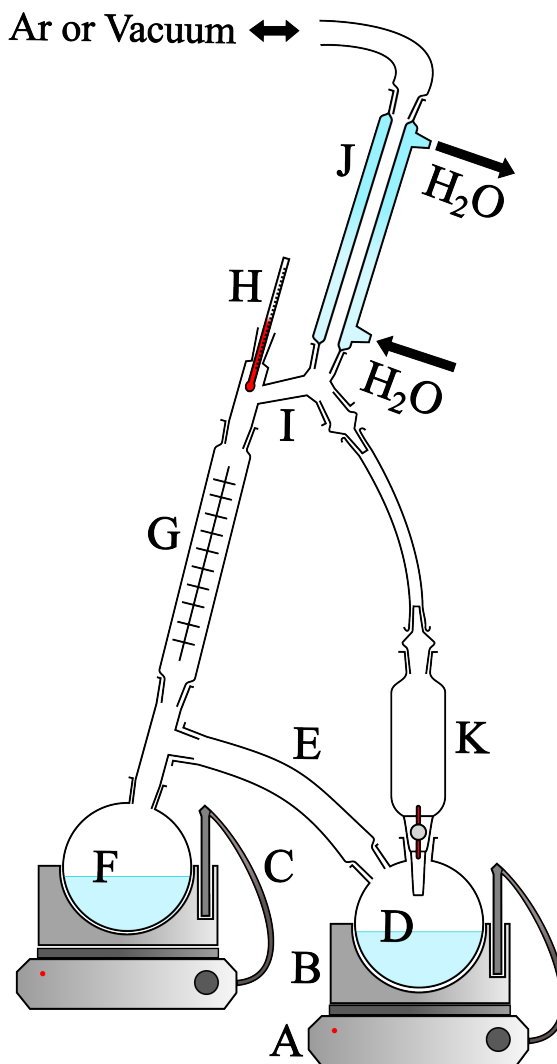


## 9 Supporting Information

Details of practical method and data analysis which cramped the presentation of material in the present paper have been gathered and elucidated within this section.

### 9.1 Notes on Synthesis of Diitaconate

Dimethyl itaconate was synthesised by esterification of itaconic acid with methanol by the Fischer method, here we describe the practical method by which this was achieved. Figure 14 shows how this reaction mixture was placed in the lower round-bottomed flask (Fig. 14D), and brought to a vigorous boil at around 130 °C. Vapours of both water and methanol passed up an insulated glass tube (Fig. 14E) and were captured in a distillation column (Fig. 14G) which was connected to a reboiler (Fig. 14F) at its base. The temperature at the top of the column (Fig. 14H) was maintained at around 70 °C by adjusting the temperature of the reboiler (120 °C) and the level of insulation applied around the column; this ensured a steady stream of methanol-rich vapours passing through the Claisen head (Fig. 14I) and re-condensing within the Liebig condenser (Fig. 14J). Methanol-rich liquid was there-with returned to the lower round-bottomed flask via a dropping funnel (Fig. 14K), whilst water was caught in the reboiler (Fig. 14F), pushing the reversible esterification reaction toward the desired diester product.



**Figure 14:** Apparatus for synthesis of dimethyl itaconate. (A) Hotplate stirrer (B) Aluminium bowl (C) Temperature controller probe (D) Three-necked round-bottomed flask containing the reaction mixture (E) Insulated glass tube (F) Single-necked round-bottomed flask acting as a distillation column reboiler containing water (G) Distillation column (H) Thermometer (I) Four-way Claisen head (J) Liebig condenser (K) Dropping funnel for recycled methanol

Before initiating the reaction by starting both the hotplate stirrers (Fig. 14A), an argon-rich atmosphere was generated by using a Schlenk line to cycle a vacuum (500 mbar) and flow of argon gas ( $350 \text{ mL min}^{-1}$ ); this was to discourage unwanted oxidation reactions and reduce the chances of vapour ignition. During the reaction however a valve was opened to allow pressure equilibration between the reaction vessels and the atmosphere in order to miti-

gate the risk of explosion. Methanol cycling was halted 3.5 h into the reaction by closing the dropping funnel (Fig. 14K), after which all heating was ceased on reaching 4 h.

Having trialled several solvents (e.g. cyclohexane, n-hexane), toluene was found to be the most suitable as its relatively higher affinity for dimethyl itaconate during liquid-liquid extraction led to acceptable yields at conclusion of workup. The final product mixture retrieved from the round-bottomed flask (Fig. 14D) was placed in a separating funnel with toluene in a 1:3 volume ratio. The separating funnel contents were washed with water three times, saturated sodium bicarbonate solution twice, and saturated sodium chloride once; each washing step involved a volume ratio of 2:1 between the previously obtained organic fraction and the next volume of washing liquid used. A gravity filtration step was then carried out between the water and sodium bicarbonate washing steps to remove insolubles. Following washing, the organic fraction was removed to an Erlenmeyer flask and dried using anhydrous magnesium sulphate; thereafter removed using gravity filtration. Dry organics were placed in a rotary evaporator and heated at 40 °C whilst the pressure was reduced from an initial 70 mbar to 20 mbar over an average period of 8 h; this removed all but a small mass of toluene and water from the liquid product. From this product, dimethyl itaconate was crystallised out at room temperature by addition of a nucleating agent. The nucleating agent was crystalline dimethyl itaconate obtained directly from a particularly pure batch of rotary evaporation product which had crystallised in a fridge at 3°C. The above method afforded a 45% yield of product, although higher values may be possible if an extended period of boiling were adopted.

## 9.2 Notes on the TLC of Diitaconates

Initial estimates of purity were obtained from spotting chemicals onto thin layer chromatography plates (polyester with a layer of silica gel and ultraviolet indicator; 6 nm pore diameter and 0.2 mm layer thickness) and using a mixture of dichloromethane and methanol (2.5:7.5 v/v respectively) as the mobile phase. On completion, spot(s) could be observed under ultraviolet light ( $\lambda = 254$  nm) and their retardation factors ( $R_F$ ) calculated using equation 21, relating this metric with the distance migrated by the substance in question ( $b$  in mm) and that travelled by the mobile phase ( $a$  in mm)<sup>75</sup>.

$$R_F = \frac{b}{a} \quad (21)$$

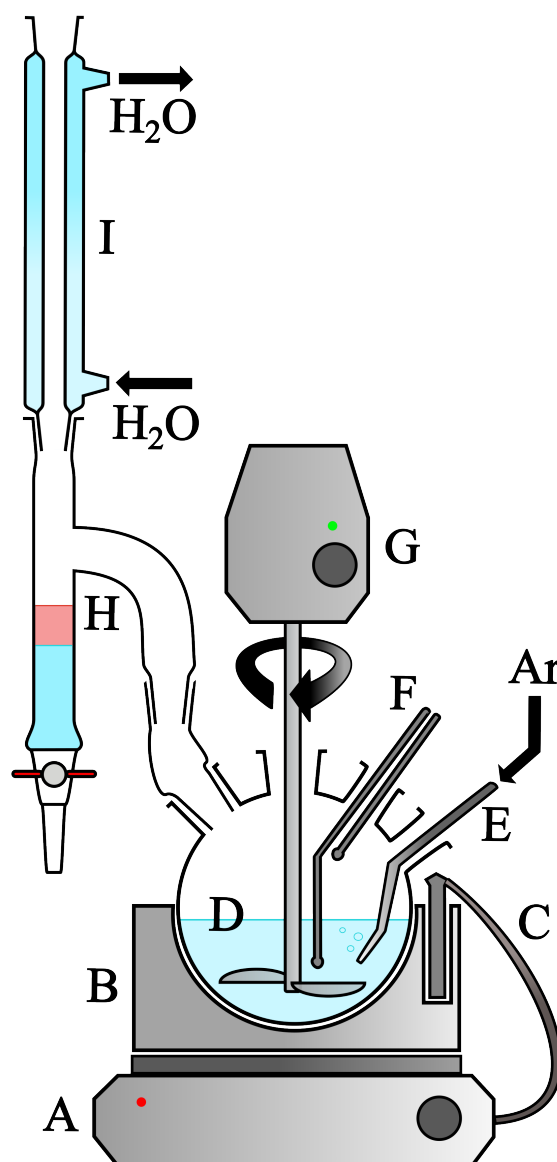
Having solvated and diluted pure itaconic acid to a 1.25% w/w solution using stock mobile phase, this solution was spotted on a plate- this produced a single spot with an  $R_F$  value of 0.79. Meanwhile, having spotted a 5% w/w solution of the final crystalline product (expected to be dimethyl itaconate), a single spot with an  $R_F$  of 0.91 was obtained. These results provided benchmarks by which reaction progress could be gauged as well as supporting the hypothesis that these two crystalline materials were reasonably pure, although the presence of non-fluorescent or trace chemicals cannot be ruled out using TLC alone.

Samples taken periodically from the reaction vessel during the course of reflux revealed a change in composition: the first sample (retrieved after 15 min of heating) presented a single spot with an  $R_F$  value of 0.81, while a final sample (retrieved after 4 h) presented two spots with  $R_F$  values of 0.90 and 0.81 respectively. Given the similarity of these samples with those of the crystalline reactant and product analysed previously, these results suggest that partial conversion of the reactant itaconic acid into the desired product dimethyl itaconate was achieved.

### 9.3 Notes on Synthesis of Unsaturated Polyester Resin

Figure 15 shows the apparatus used to synthesise the unsaturated polyester. A Dean-Stark trap (Fig. 15I) was used to allow the ‘azeotropic method’ of polyester synthesis (as opposed to the ‘dry’ method) which reduced the time taken to synthesise a high molecular weight polyester<sup>56</sup>. In this method, an entraining agent (mixed xylenes) which can form an azeotrope with the water was added to the mixture (5% w/w)<sup>58,76</sup>. This increases the rate at which water is condensed and captured by the Dean-Stark trap, driving the reaction forward by removing the unwanted byproduct of water from the reaction mixture. Furthermore, argon gas was used to sparge the reaction mixture (Fig. 15E) facilitating the removal of water from the bulk mixture and mitigating the combustion risk associated with high temperature xylenes<sup>56</sup>.

The synthesis procedure was initiated by charging the four-necked round-bottomed flask (Fig. 15D) with the requisite masses of the two monomers and xylenes, opening the argon taps to provide a flow of sparging gas at the rate of 5 bubbles per second through the argon injector (Fig. 15E)<sup>56</sup>, opening the water taps to actively cool the Liebig condenser (Fig. 15I), setting the overhead stirrer to a speed of 50 rpm (Fig. 15G)<sup>56</sup>, and bringing the reaction mixtures temperature (Fig. 15F) up to 160°C using the hotplate (Fig. 15A). During the procedure the mixture’s temperature was raised monotonically by 10°C h<sup>-1</sup> until it reached a maximum of 210°C, at which it was held.



**Figure 15:** Apparatus for synthesis of unsaturated poly(propylene itaconate). (A) Hotplate stirrer (B) Aluminium bowl (C) Temperature controller probe (D) Four-necked round-bottomed flask containing the reaction mixture (E) Stainless-steel argon injector (F) Probes for batch and vapour temperatures (G) Overhead stirrer (H) Dean-Stark trap (I) Liebig condenser.

The reaction progress was ascertained by periodically retrieving a sample of the reaction mixture and ascertaining its acid number, in units of mg g<sup>-1</sup>, describing the amount of potassium hydroxide required to neutralise one gram of sampled unsaturated polyester. This approximate form of end-group analysis is commonly used in unsaturated polyester synthesis<sup>42,43,45,56</sup> and provides a proxy value estimating the number of un-esterified acid groups remaining within

the unsaturated polyester. An American Society for Testing and Materials standard ("ASTM D4662-20") was used for this purpose. In-line with common practice, heating of the polyesterifying mixture was ceased at an acid value of 50<sup>56,58,76</sup>. On cessation of heating, an outlet was opened for egress of remnant xylenes and the unsaturated polyester was allowed to cool in an air bath.

Since the viscosity of a cooling unsaturated polyester will increase monotonically until it reaches room temperature, where it exists as a solid thermoplastic mass, it is common to prepare the unsaturated polyester resin immediately after cessation of heating by adding reactive diluent directly to the cooling unsaturated polyester<sup>56</sup>. Unfortunately, reactive diluents are liable to polymerise during this addition causing a premature gelation. Mitigation against hardening into thermoplastic or premature gelation into rubber can be achieved by addition of an addition-polymerisation inhibiting agent at a suitable intermediate temperature<sup>25,56,57</sup>. In this work, we awaited a temperature of 100 °C before adding the below specified amounts of inhibitor and reactive diluent.

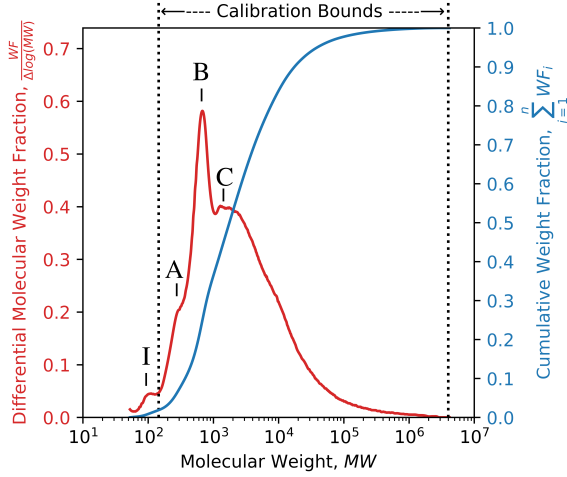
Here, we used the commonly-used free-radical scavenger hydroquinone as an inhibiting agent, which was added so as to make up 0.06% w/w of the cooling unsaturated polyester mixture. Hydroquinone was dissolved in a small amount of propylene glycol before addition to ensure homogenous mixing into the bulk<sup>56</sup>. The inhibitor percentage was determined through a process of trial-and-error, where lower percentages tended to facilitate premature gelation of unsaturated polyester resins on the shelf, whilst higher percentages entirely prevented the intended curing procedures for these resins. The percentage of inhibiting agent was above that which is commonly used in the synthesis of conventional unsaturated polyester resins (approximately 0.02%)<sup>56,57</sup>. This was a consequence of both the relatively higher

proportion of unsaturated sites along the polyester backbone and a higher copolymerisation affinity between the reactive diluent and the sites of unsaturation.

Following introduction and homogenisation of inhibiting agent into the unsaturated polyester mixture, the reactive diluent (dimethyl itaconate) was added such that it made up 30% w/w of the final unsaturated polyester resin.

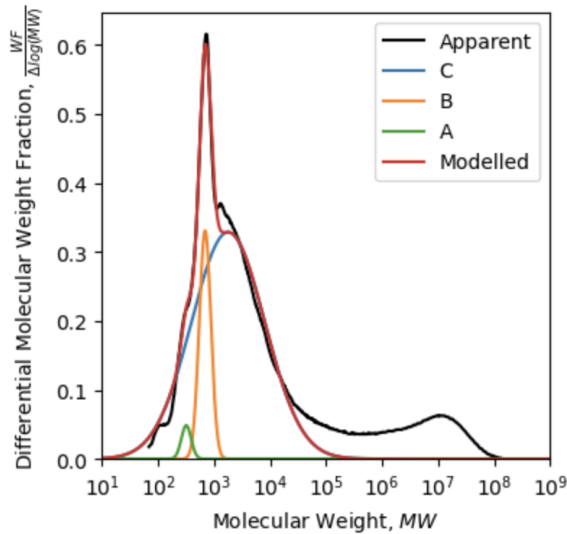
## 9.4 Notes on Gel Permeation Chromatography

The following reconstruction and deconvolutions are exploratory sensitivity analyses only; they are not used to report measured molar-mass averages or to establish molecular structures. In a separate experimental series (not considered in the context of the present work), a similar polyester was synthesised with a reduced content of itaconate moieties in its backbone. This was achieved by replacing some of the itaconic acid used during synthesis with succinic acid, producing poly(propylene succinate-co-itaconate). Figure 16 depicts the GPC results obtained on analysing this alternative formulation. On comparison with the GPC results obtained for poly(propylene itaconate) as detailed in the present work it becomes apparent that this alternative formulation lacked the prominent early-eluting high-hydrodynamic-volume feature seen for the present poly(propylene itaconate) samples. As the closest approximation to the distribution associated a poly(propylene itaconate) polyester that lacked the prominent early-eluting high-hydrodynamic-volume feature, we re-scaled the tail section of Figure 16 and inserted this into the apparent distribution for poly(propylene itaconate) presented in the present work. This allowed the calculation of both number and weight average molecular weights for the hypothetically uncrosslinked polyester.



**Figure 16:** A graph depicting the distribution of mass across various polymer chain lengths for an alternative polyester sample tested using GPC.

An alternative line of investigation undertaken with respect to the GPC of these polyesters was that associated with the deconvolution of the apparent distribution. By identification of three physical peaks (Figure 4A/B/C) mathematical deconvolution was undertaken by fitting a log normal distribution to the three curves in isolation (by cropping the apparent distribution suitably), before fitting the three in unison (in an additive fashion) to the apparent distribution. Figure 17 shows the three deconvoluted peaks and their fit relative to the apparent distribution.



**Figure 17:** A graph of all three deconvoluted peaks which were identified from within the apparent distribution.

## 9.5 Notes on Mechanical Test Data Processing

Here is an unabridged overview of all the data processing undertaken during the mechanical testing of samples.

**Euler Buckling** To mitigate against concerns of Euler buckling taking place in samples of the proposed dimensions, we calculated the critical load based on Euler's critical load equation and some estimated properties for conventional polyester thermosets<sup>64</sup>. Equation 22 shows how the critical load ( $P_{cr}$  in N) is related to the elastic modulus of the material ( $E$  in Pa), the second moment of area ( $I$  in  $m^4$ ), the effective length factor ( $K$ , unitless), and the initial length of the column in question ( $L$  in m).

$$P_{cr} = \frac{\pi EI}{(KL)^2} \quad (22)$$

Apparent compressive moduli for conventional thermoset polyesters were expected to range from 2-4.5 GPa<sup>24</sup>. The second moment of the area for the top surface of the cylindrical sample with radius 6.35 mm was calculated to be  $1.276 \times 10^{-9} m^4$  by assuming it to be a perfect filled circular area, for which the generic equation 23 applied (where the  $r$  is the radius of the cylinder in m).

$$I_y = \frac{\pi r^4}{4} \quad (23)$$

By assuming that both ends of the column would be braced against rotation or translation (given their compression between two platens), the theoretical column effective length factor  $K$  would have been 0.5; as a further precaution a conservative design value of 0.65 was assumed based on pertinent literature<sup>77</sup>. On these assumptions it was estimated that the samples in question would be likely to undergo Euler buckling at an applied load of between 433-976 kN. As the compressive strength expected of conventional unsaturated polyester thermosets

are expected to fall within the range of 89-252 MPa, corresponding to maximum compressive forces of between 11-32 kN (given the surface area of the samples), we concluded that the samples were unlikely to experience Euler buckling.

**Correction of Raw Data** Finite compliance of the testing machine introduced an additional displacement contribution into raw exported datasets. To correct for this, a blank compression test was performed with no sample positioned between the platens, generating a force-displacement curve corresponding to the inherent compliance of the machine. This curve was subsequently used as a correctional dataset. Equation 24 shows how the corrected displacement value ( $d$  in mm) was obtained by taking a raw displacement value ( $d_{\text{raw}}$  in mm) and subtracting the corresponding correctional displacement value ( $d_{\text{cor}}$ ).

$$d = d_{\text{raw}} - d_{\text{cor}} \quad (24)$$

**Processing of Corrected Data** Load values from a sample's corrected dataset were converted into engineering stress values by dividing by the initial surface area. Each cylindrical sample's diameter was measured using a "Magnusson" digital caliper (accuracy  $\pm 0.02$  mm) from three different positions ( $\phi_1, \phi_2, \phi_3$  in mm), each offset by roughly  $+45^\circ$  around the circumference, and the mean of these values ( $\phi_{\text{avg}}$  in mm) calculated (see Equation 25).

$$\phi_{\text{avg}} = \frac{\phi_1 + \phi_2 + \phi_3}{3} \quad (25)$$

Equation 26 shows how the initial top-face surface area was calculated from the average top surface diameter  $\phi_{\text{avg}}$ .

$$A_{\text{init}} = 0.25 \times \pi \times \phi_{\text{avg}}^2 \quad (26)$$

Subsequently, the engineering stress ( $\sigma$  in MPa) could be calculated (Equation 27) by

considering the force applied ( $F$  in N) and the initial top surface area,  $A_{\text{init}}$ <sup>65</sup>.

$$\sigma_{\text{eng}} = \frac{F}{A_{\text{init}}} \quad (27)$$

Displacement values from a sample's corrected dataset were converted into engineering strain by taking the difference between the initial length of the samples and the length of the sample during compression. The initial length ( $l_{\text{init}}$  in mm) of sample was calculated by taking the mean of three length measurements ( $l_1, l_2, l_3$  in mm) (these taken in the same fashion as for diameter) as shown in Equation 28.

$$l_{\text{init}} = \frac{l_1 + l_2 + l_3}{3} \quad (28)$$

Equation 29 shows the simple link between length of a sample at a given point during a compressive test ( $l$  in mm), the initial sample length ( $l_{\text{init}}$ ) and the corrected displacement at that point  $d$ .

$$l = l_{\text{init}} - d \quad (29)$$

Subsequently, the engineering strain was calculated by finding the absolute change in sample length ( $\Delta l_{\text{init}}$  in mm) and dividing this by the initial length of the sample ( $l_{\text{init}}$ ) as shown in Equation 30.

$$\epsilon_{\text{eng}} = \frac{\Delta l_{\text{init}}}{l_{\text{init}}} = \frac{l - l_{\text{init}}}{l_{\text{init}}} \quad (30)$$

**Stress-Strain Smoothing** To produce a smoothed dataset (from which apparent compressive modulus and peak compressive strain could be deduced) a combination of sample-rate reduction and filtering were applied to downsample the stress-strain dataset. Downsampling was achieved using a windowed averaging technique, whereby the unsmoothed dataset ( $p_{\text{us}}$ ) was split into a number of windows along the strain-axis ( $p_{\text{s}}$ ) by electing a suitable number of points per window ( $p_{\text{w}}$ ), as shown in Equation 31. For convenience,  $p_{\text{us}}$  is required to be divisible by  $p_{\text{w}}$ , and  $p_{\text{s}} \geq 200$  was found to give reliable results. Given the constant size of the unsmoothed dataset ( $p_{\text{us}} = 1,000$ ), a

constant number of points per window was elected ( $p_w = 5$ ).

$$p_s = p_{us}/p_w \quad (31)$$

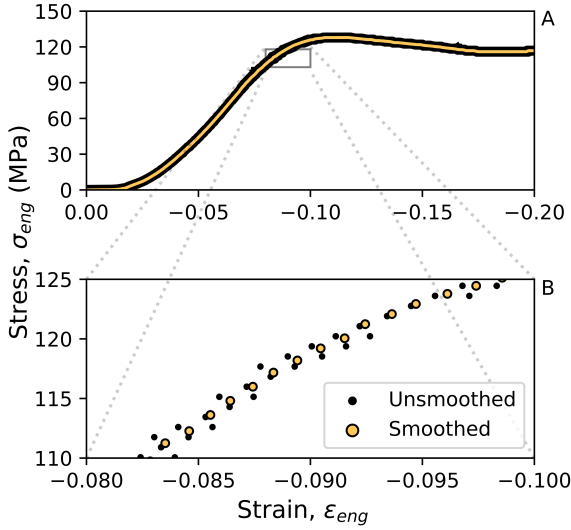
Equation 32 shows how the process of averaging stress ( $\sigma_{eng}^i$ ) and strain ( $\epsilon_{eng}^i$ ) values within a given window yielded a single point described by its average stress ( $\bar{\sigma}_{eng}^j$ ) and strain ( $\bar{\epsilon}_{eng}^j$ ).

$$\bar{\sigma}_{eng}^j, \bar{\epsilon}_{eng}^j = \frac{\sum_{i=1}^{p_w} \sigma_{eng}^i}{p_w}, \frac{\sum_{i=1}^{p_w} \epsilon_{eng}^i}{p_w} \quad (32)$$

Averaged points from each window were assembled to form the new downsampled dataset depicted in Equation 33.

$$[[\bar{\sigma}_{eng}^j, \bar{\epsilon}_{eng}^j], [\bar{\sigma}_{eng}^{j+1}, \bar{\epsilon}_{eng}^{j+1}], \dots, [\bar{\sigma}_{eng}^{p_s}, \bar{\epsilon}_{eng}^{p_s}]] \quad (33)$$

Figure 18 demonstrates the efficacy of this technique in removing noise from the unsmoothed stress-strain dataset.



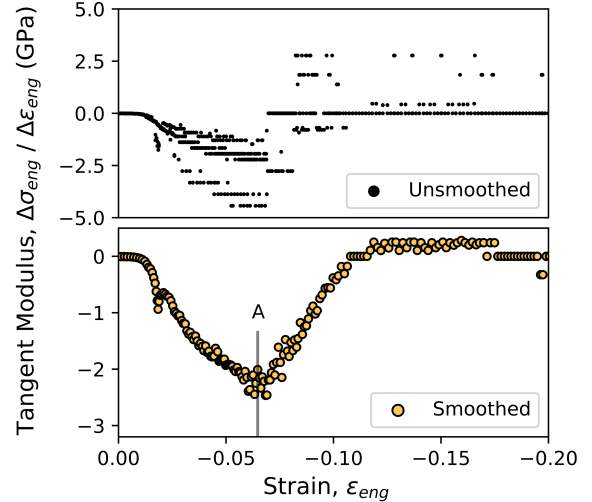
**Figure 18:** Multi-scale depiction of the stress-strain dataset before and after downscaling. (A) Overview line plot. (B) Zoomed scatter plot.

**Derivative Analysis** The proportional limit is defined as the stress at which a material ceases to obey Hooke's law (i.e. stress no longer increases proportionally with strain). The gradient of the preceding elastic region of deformation is generally defined as the apparent compressive modulus. However, given that the stress-strain curves did not consistently exhibit a linear region, the limit of proportionality was equated to the point

at which the stress-strain curve began to level off (i.e. the first point of inflection). To find this inflection point, the first derivative of stress with respect to strain (i.e. tangent modulus  $E_t$  in MPa) was calculated from the smoothed stress-strain dataset by taking into account the instantaneous change in both stress ( $\Delta\sigma_{eng}$ ) and strain ( $\Delta\epsilon_{eng}$ ), and given by Equation 34.

$$E_t = \frac{\Delta\sigma_{eng}}{\Delta\epsilon_{eng}} = \frac{\sigma_{eng}^{j+1} - \sigma_{eng}^j}{\epsilon_{eng}^{j+1} - \epsilon_{eng}^j} \quad (34)$$

Figure 19 shows how smoothing of the stress-strain dataset facilitated the analysis of the resultant tangent modulus versus strain dataset. Step-like features are apparent in the tangent-modulus curve obtained from the unsmoothed stress-strain data. These may be consistent with either distretisation or internal processing of the exported displacement signals and subsequently amplified by the process of numerical differentiation. To avoid undue physical significance being assigned to these artefacts, proportional-limit detection was performed upon a smoothed stress-strain dataset.



**Figure 19:** A comparison of the tangent modulus vs strain curves obtained from both the raw and the smoothed counterpart stress-strain datasets. (A) Strain at first inflection point.

Detection of the first inflection point was automated using an algorithm that iterated chronologically through the tangent modulus versus strain dataset from low to high strains. At each point, the algorithm cal-



culates an average gradient exhibited by an equal number of points both before and after the current strain value (the ‘point horizon’). Using a point horizon of ten, the first inflection point was defined as the first point at which the summed forward and backward gradients fell below zero. This point was taken as the proportional limit, as shown in Figure 19A.

## 9.6 Notes on Gaussian Processes

Figure 20 presents a matrix of slice plots involving the two Gaussian processes as they existed at the culmination of the multi-objective experiment. Every plot holds two-thirds of the predictors constant whilst the column dictates which of the three predictors is varied. Each row of the matrix conditions the Gaussian process  $\mathcal{GP}$  on a different set of predictor constants, these are derived from either the 14<sup>th</sup> sample ( $\mathbf{x}^{i=14}$ ), the 68<sup>th</sup> sample ( $\mathbf{x}^{i=68}$ ), or the mean of all samples’ predictors ( $\mathbf{x}^\mu$ ). The graphs are further annotated with vertical lines representing sample predictor values ( $\{x_1, x_2, x_3\}$ ):  $\mathbf{x}^{i=14} = \{0.98, 0.5, 0.93\}$   $\mathbf{x}^{i=68} = \{0.72, 0.4, 0.5\}$   $\mathbf{x}^\mu = \{0.71, 0.64, 0.64\}$ .

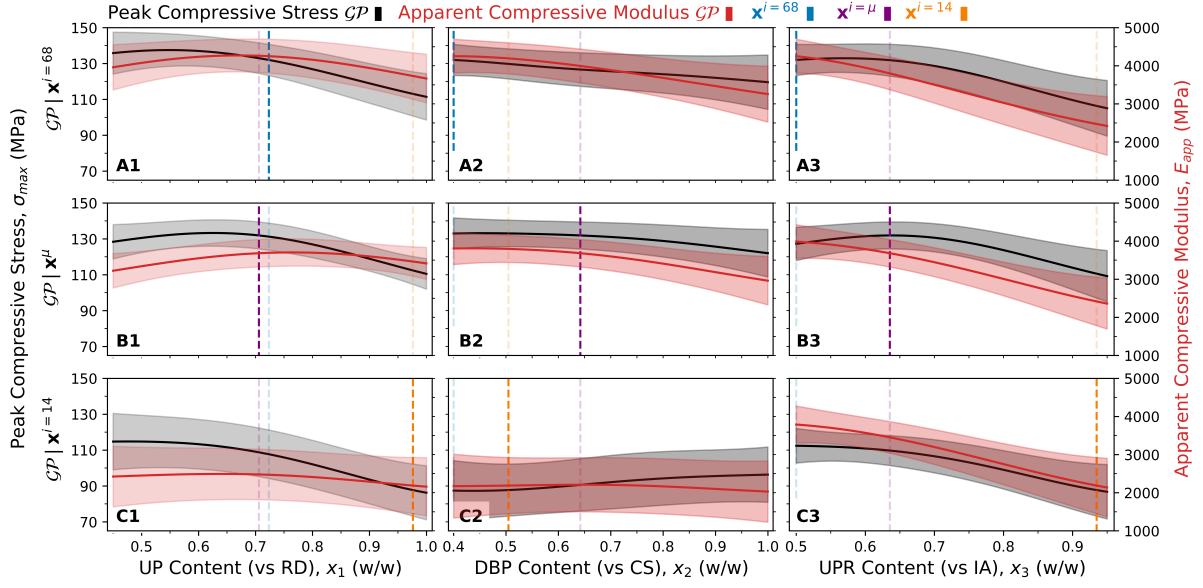
Row B of Figure 20 describes the processes as conditioned upon the experimental mean of sample predictors ( $\mathcal{GP}|\mathbf{x}^\mu$ ); these graphs provide rough intuition with respect to the objective topography of the parameter space but naturally fail to account for why specific combinations of predictor values (e.g. sample 68 or 14) exhibit higher or lower performance in both the metrics of interest.

In Figure 20A we see the processes conditioned on the 68<sup>th</sup> sample’s predictors  $\mathcal{GP}|\mathbf{x}^{i=68}$ : in the case of predictor  $x_3$ ’s peak compressive stress process (Fig.20:A3) we see a topographical deviation from the mean conditioned process (Fig.20:B3). This deviation was a flattening in the region where  $x_3 = [0.5, 0.65]$  which shifted the local optimum toward  $x_3 = 0.5$ : this resulted in

an alignment of both local optima of the two processes, which partly explains why marginalisation of modified resin content by sample 68 leads to improved performance in both objective metrics.

Although the troughs apparent in the mean processes with respect to  $x_1$  and  $x_3$  (Fig.20:B1&B2) hinted at why the 14<sup>th</sup> sample’s predictor value combination produced suboptimal performance in both the objective metrics, the  $x_2$  topography failed to do so. On examination of the  $x_2$  topography of the processes conditioned on the 14<sup>th</sup> sample’s predictors  $\mathcal{GP}|\mathbf{x}^{i=14}$  we find large deviations from the mean processes: both the topographies have fallen noticeably, the apparent compressive modulus process has flattened, and the peak compressive stress process inverted such that the local optimum shifted from  $x_3 = 0.4$  to  $x_3 = 1$ . Furthermore, this inversion presented a local trough in the vicinity of the 14<sup>th</sup> sample explaining further why this sample behaved suboptimally.

These three observations are respectively reflected in the Gaussian process topographies presented in Figure 20: when considering predictor  $x_3$  (column 3: controlling balance of modified resin vs initiator/filler premix) we see that Gaussian processes clearly describe a negative correlation between  $x_3$  and optimality in all three slice plots (A3,B3,C3). These strong correlations and coinciding local optima between  $\mathcal{GP}$ ’s would explain why there is a clear global link between the balance of modified resin content and initiator/filler-premix and sample optimality, With respect to  $x_2$  (column 2: controlling initiator/filler-premix composition) the  $\mathcal{GP}$ ’s display a negative correlation between  $x_2$  and optimality in two of the slice plots (A2,B2), but this relation disappears and is replaced with a flattened and lowered process in at least one region of the parameter space (C2). In the case of  $x_1$  (column 1: controlling modified resin composition) we can see that the  $\mathcal{GP}$ ’s display a function shape with singular peaks



**Figure 20:** Slice plot of the Gaussian processes developed by the multi-objective experiment with mean function and a 95% confidence interval ( $1.96\sigma$ ) presented. UP, Unsaturated polyester. RD, Reactive diluent. DBP, Dibenzoyl peroxide. CS, Calcium sulphate. UPR, Unsaturated polyester resin. IA, Initiating agent.

at varying positions in all the different slice plots (A1,B1,C1); this added optimisation complexity explains well why there is not obvious global trend relating the modified resin content with sample optimality.